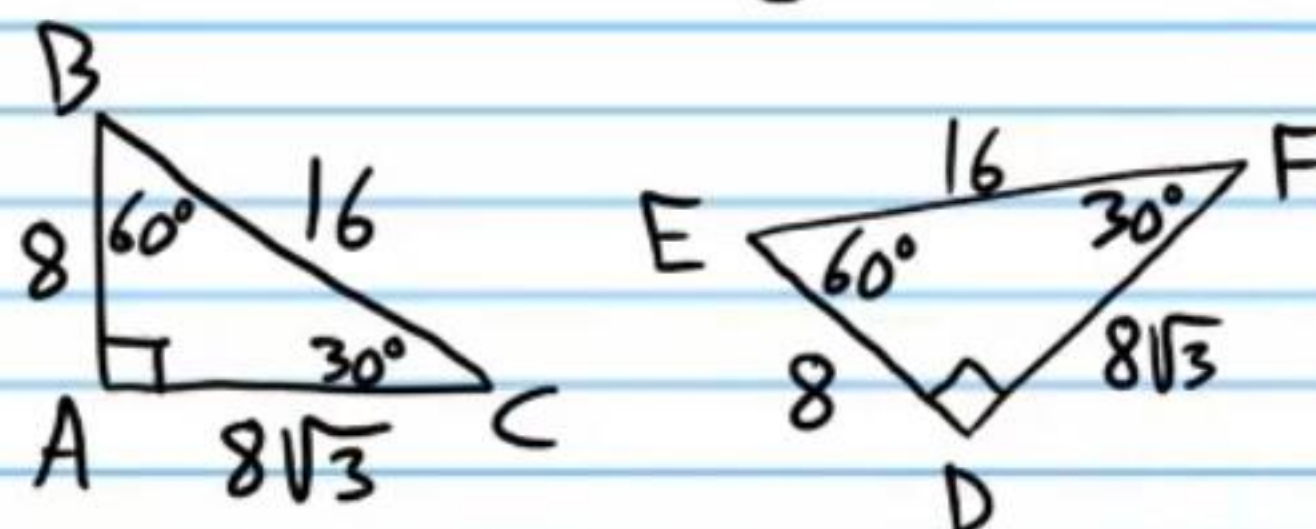


Triangle Congruence Postulates and Theorems

Congruent Polygons: Polygons are congruent if they have the same size and shape. More specifically, they are congruent if corresponding angles are congruent and corresponding sides are congruent.

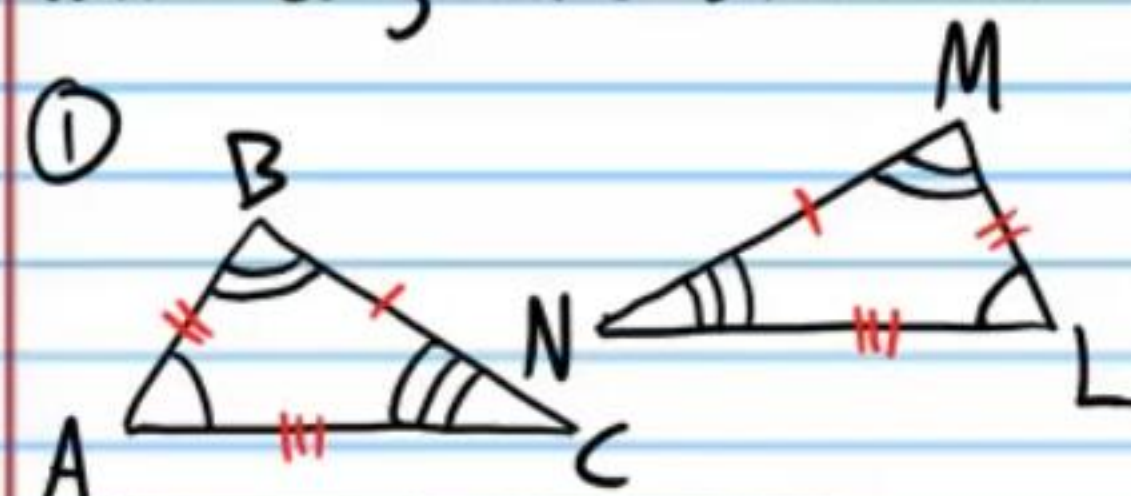
Example:



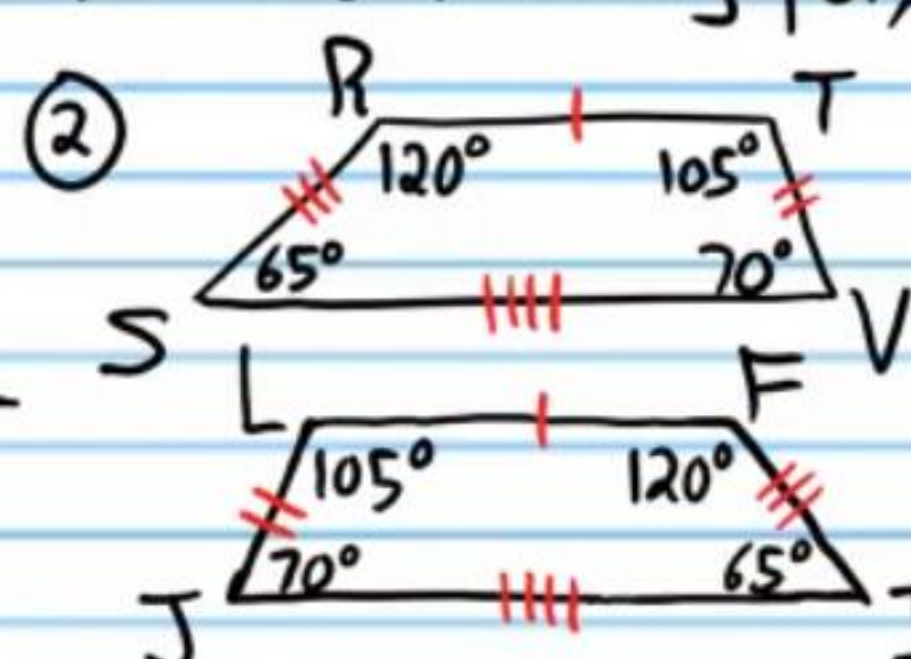
$$\triangle ABC \cong \triangle DEF$$

Congruence Statement: A mathematical expression using shapes, letters, and a congruence symbol to show that two things are congruent. The letters must be in the proper order.

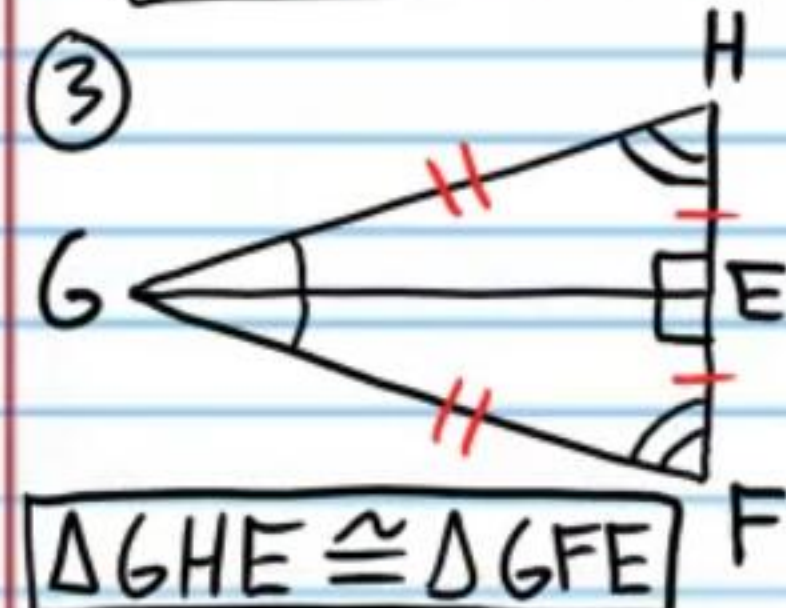
Write congruence statements for the following polygon pairs.



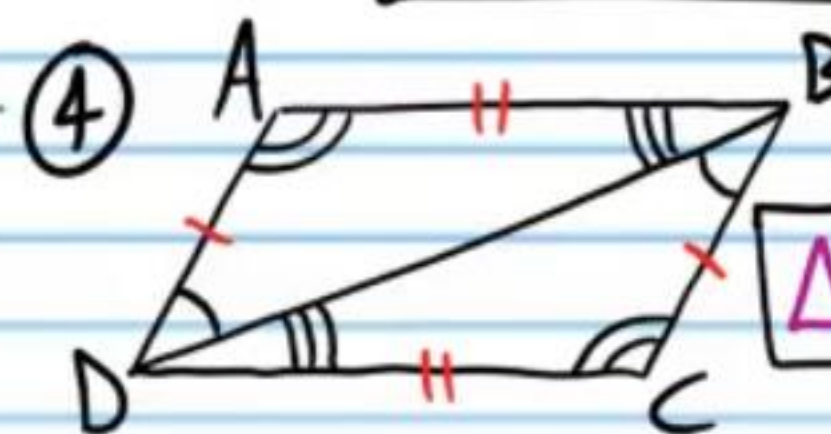
$$\triangle ABC \cong \triangle LMN$$



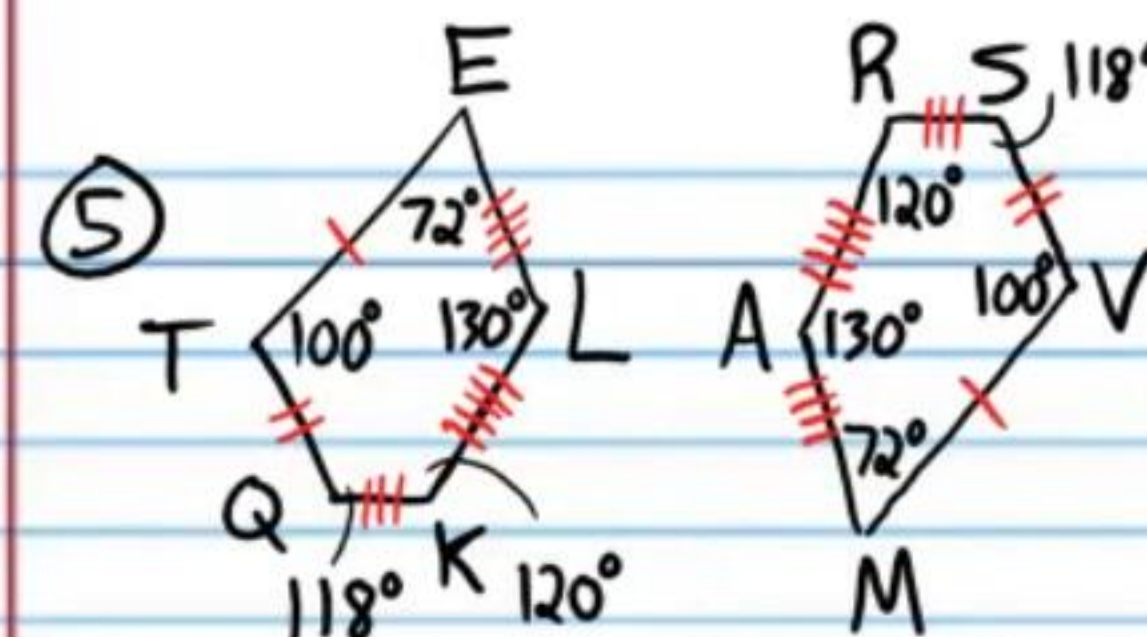
$$\square SRTV \cong \square IF LJ$$



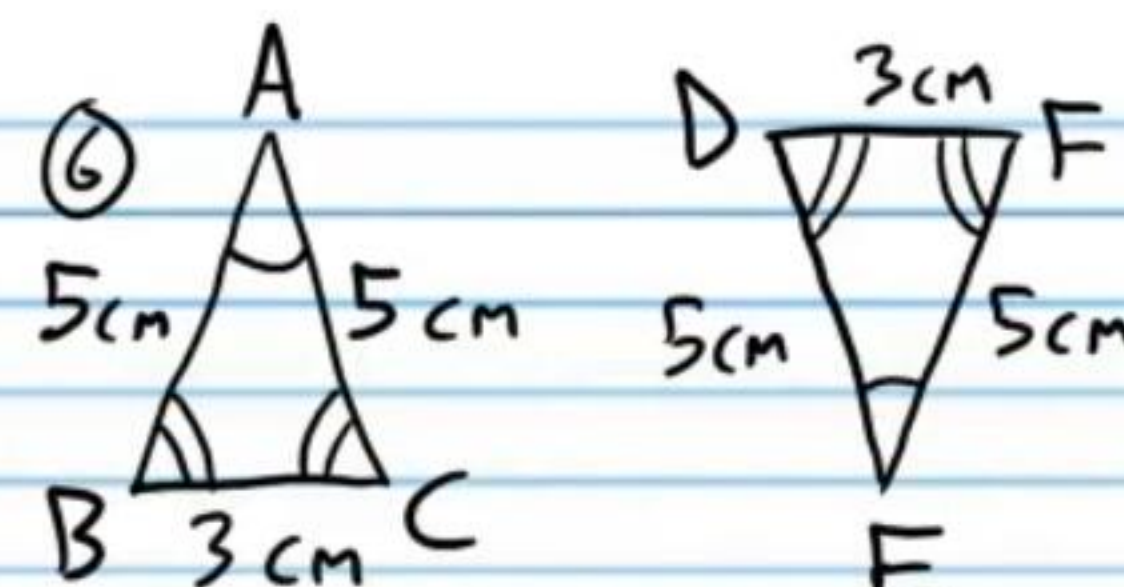
$$\triangle GHE \cong \triangle GFE$$



$$\triangle ABD \cong \triangle CDB$$



$$\square TELKQ \cong \square VMARS$$



$$\triangle ABC \cong \triangle DEF$$

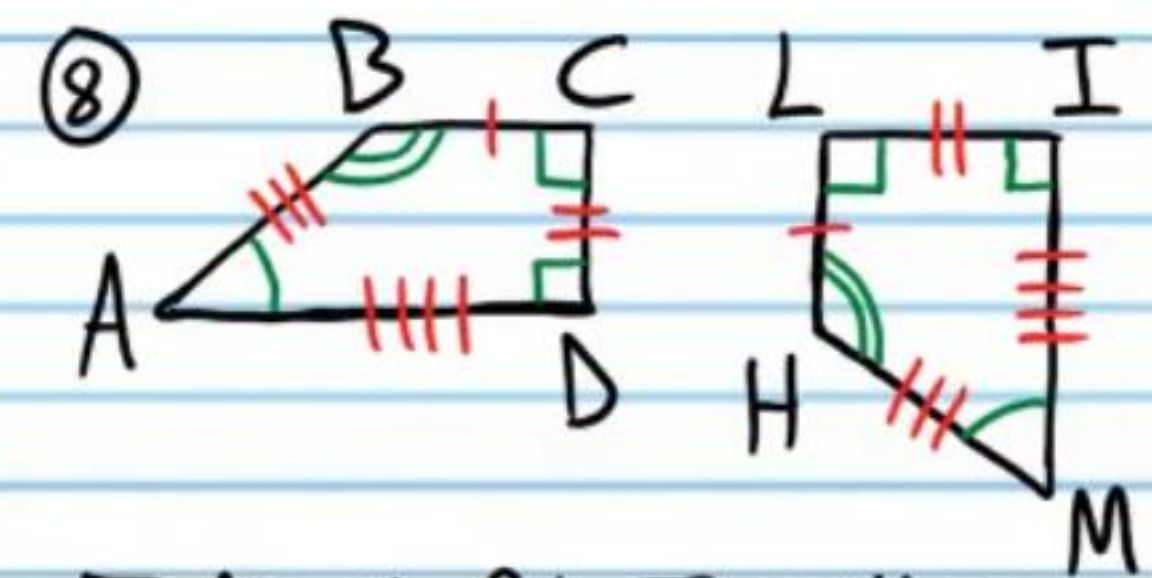
Given the polygon pairs below, rectangle the congruence statement that is not true.



$$\triangle ABC \cong \triangle KHE$$

$$\triangle ACB \cong \triangle EHK$$

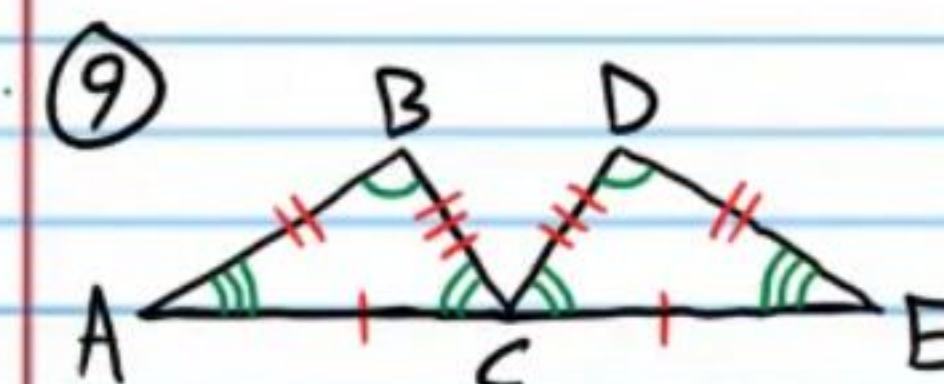
$$\triangle CAB \cong \triangle EKH$$



$$\square ABCD \cong \square MHLI$$

$$\square DCBA \cong \square ILMH$$

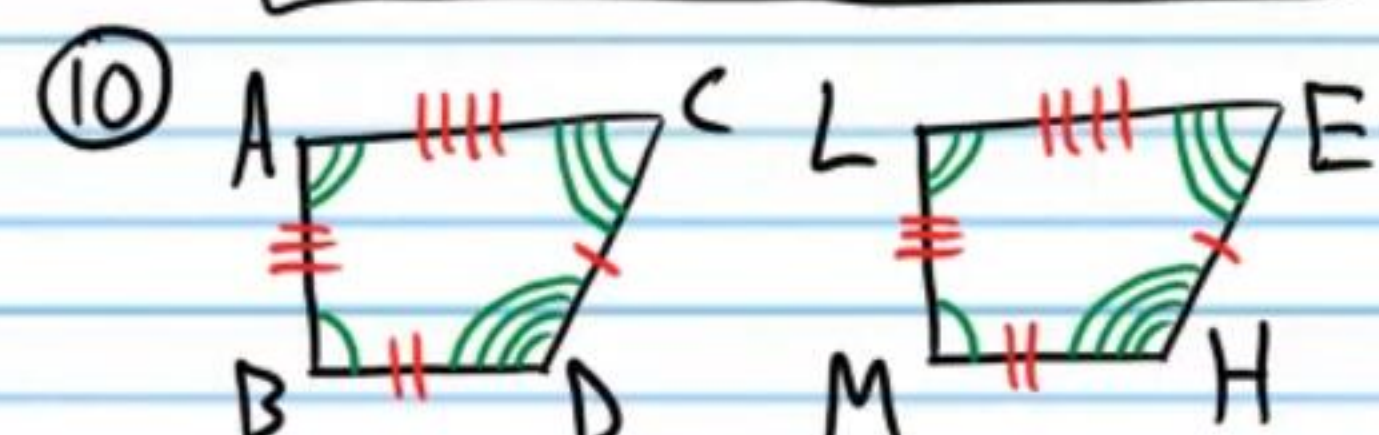
$$\square BADC \cong \square HLIM$$



$$\triangle ABC \cong \triangle ECD$$

$$\triangle CBA \cong \triangle CDE$$

$$\triangle BAC \cong \triangle DEC$$



$$\square ABDC \cong \square LMHE$$

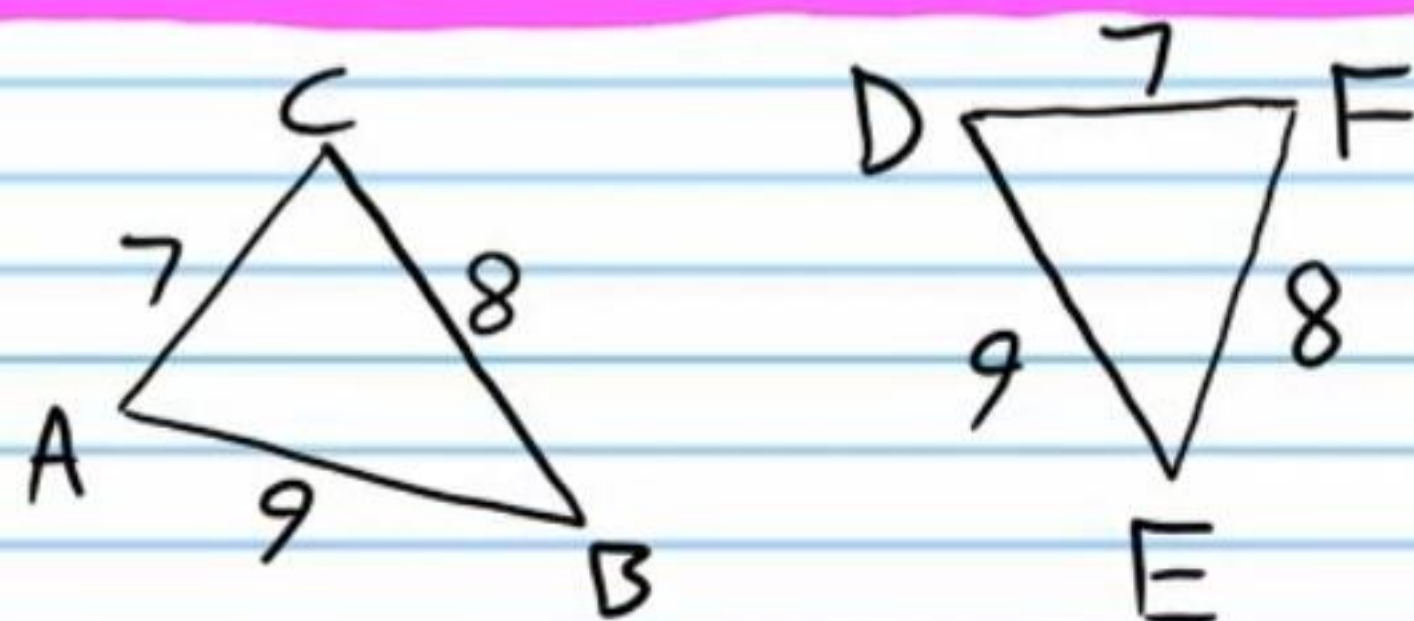
$$\square ACDB \cong \square LEHM$$

$$\square CDBA \cong \square EHL M$$

Triangle Congruence

Side-Side-Side Congruence Postulate: If each side in one triangle is congruent to each corresponding side in another triangle, then the triangles are congruent. This postulate is abbreviated SSS.

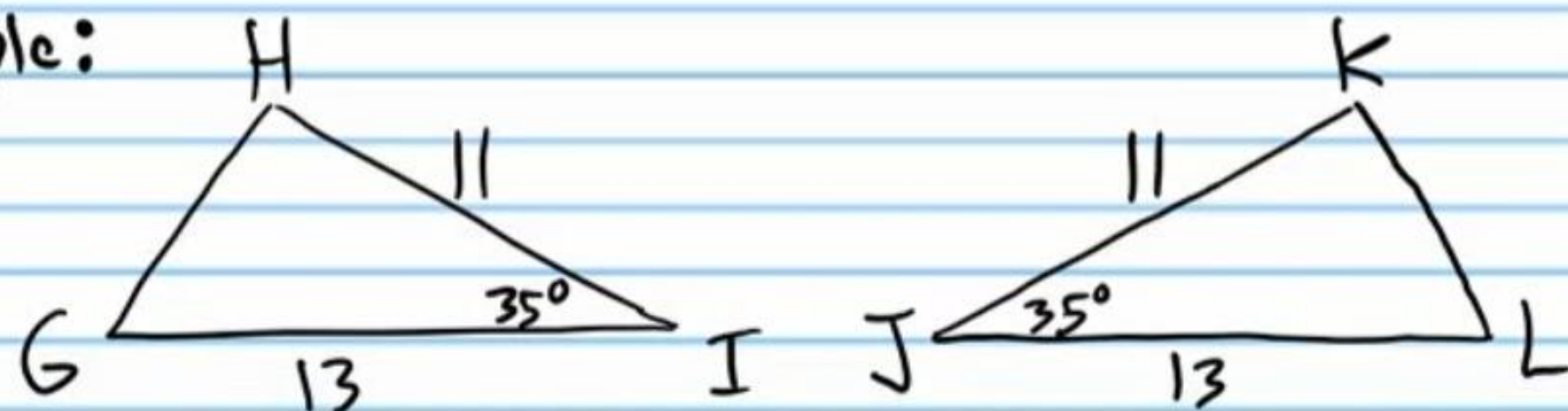
Example:



By the SSS Congruence Postulate, $\triangle ABC \cong \triangle DEF$.

Side-Angle-Side Congruence Postulate: If two sides and the included angle in one triangle are congruent to two sides and the included angle in another triangle, then the triangles are congruent. This postulate is abbreviated SAS.

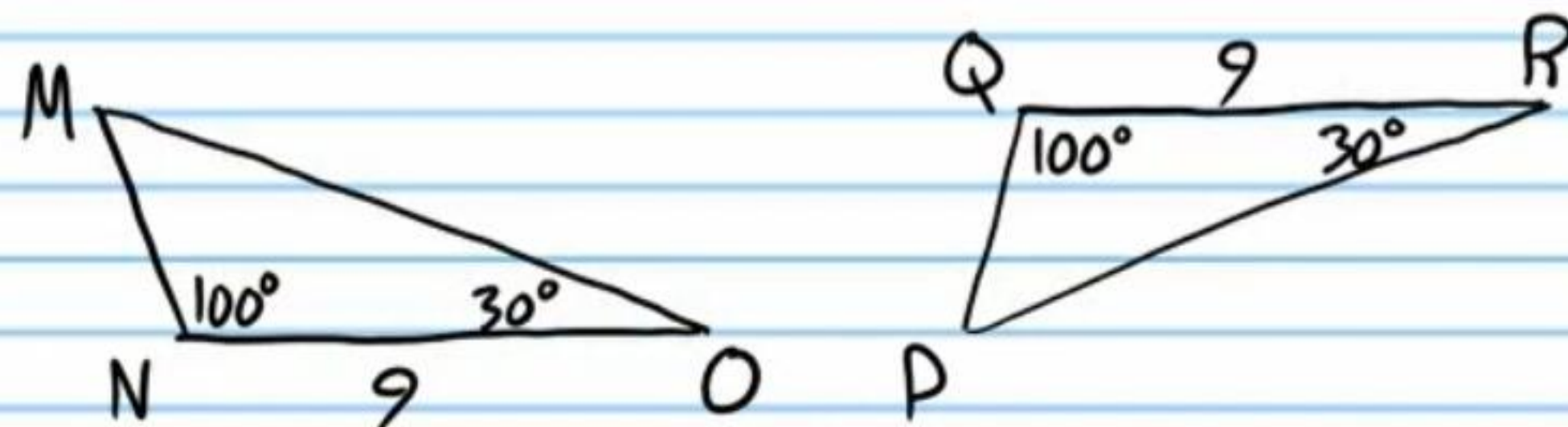
Example:



By the SAS Congruence Postulate, $\triangle GHI \cong \triangle LKJ$.

Angle-Side-Angle Congruence Postulate: If two angles and the included side of one triangle are congruent to two angles and the included side of another triangle, then the triangles are congruent. This postulate is abbreviated ASA.

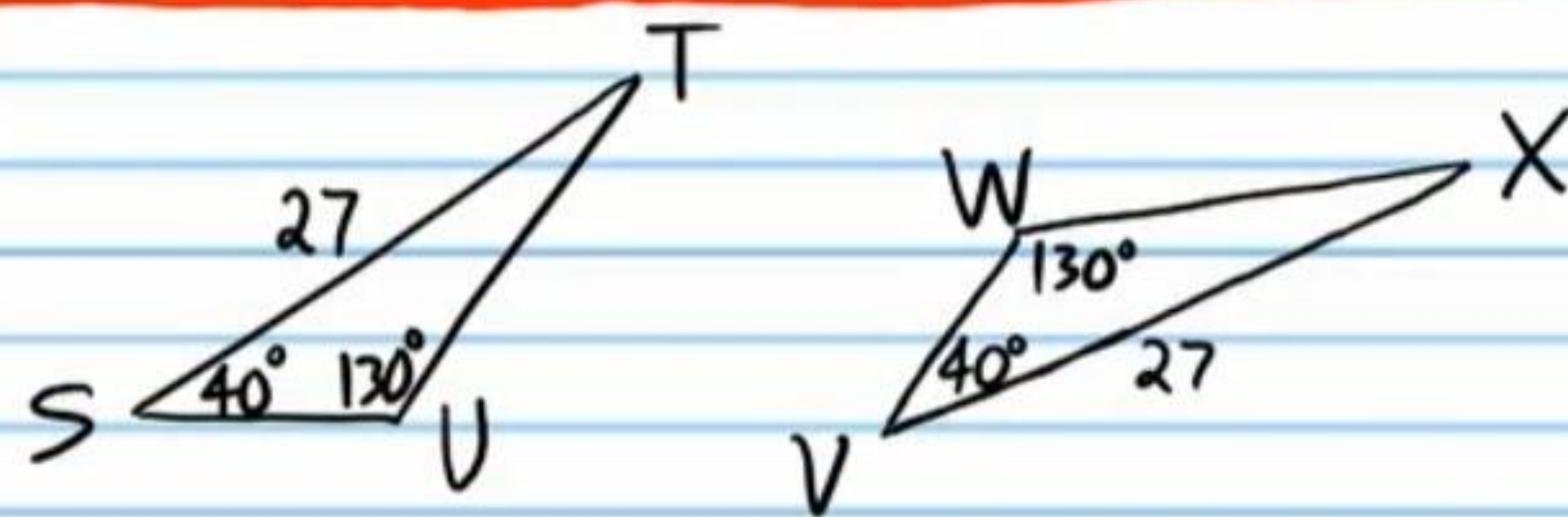
Example:



By the ASA Congruence Postulate, $\triangle MNO \cong \triangle PQR$.

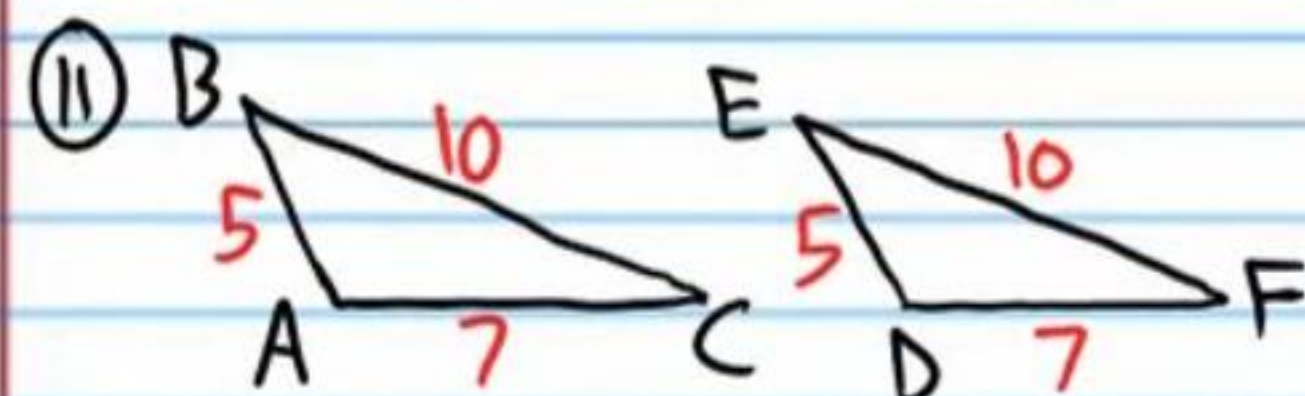
Angle-Angle-Side Congruence Theorem: If two angles and a non-included side in one triangle are congruent to two angles and a non-included side in another triangle, then the triangles are congruent. This theorem is abbreviated AAS.

Example:

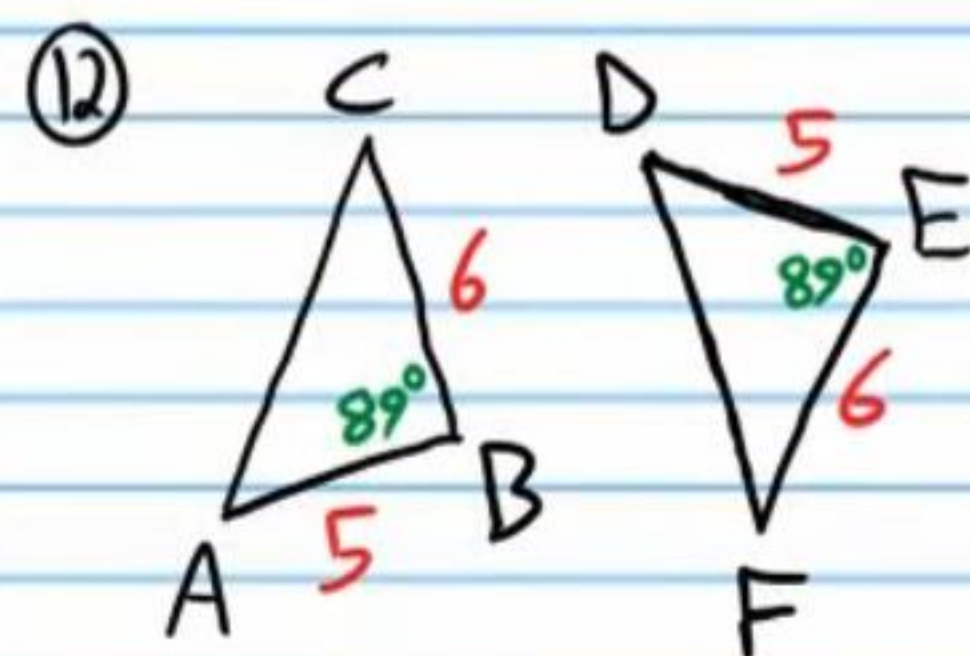


By the AAS Congruence Theorem, $\triangle SUT \cong \triangle VWX$.

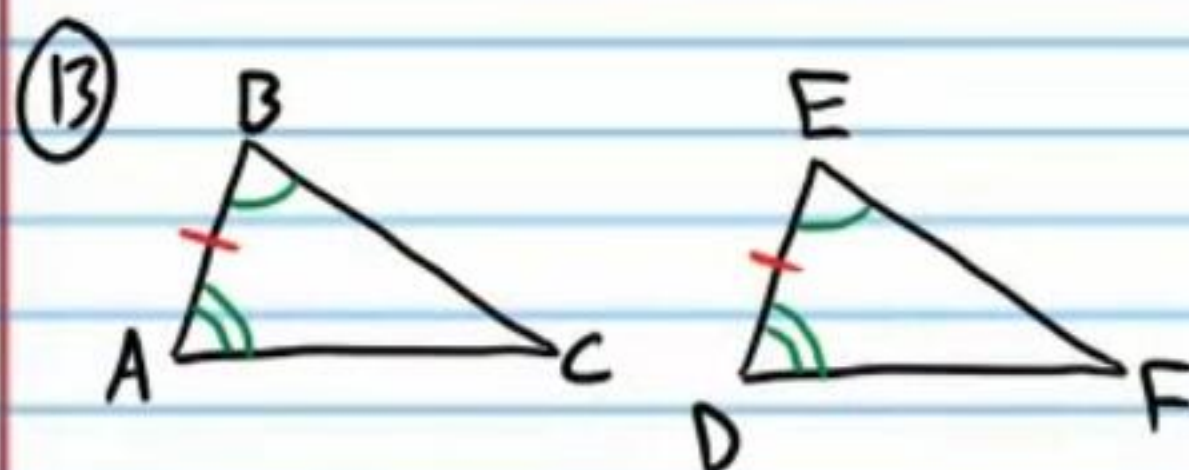
In the problems below, determine if the triangles are congruent. If they are congruent, write a proper congruence statement and the triangle postulate or theorem that proves that they are congruent. You do not have to write a formal proof (i.e. a two-column proof).



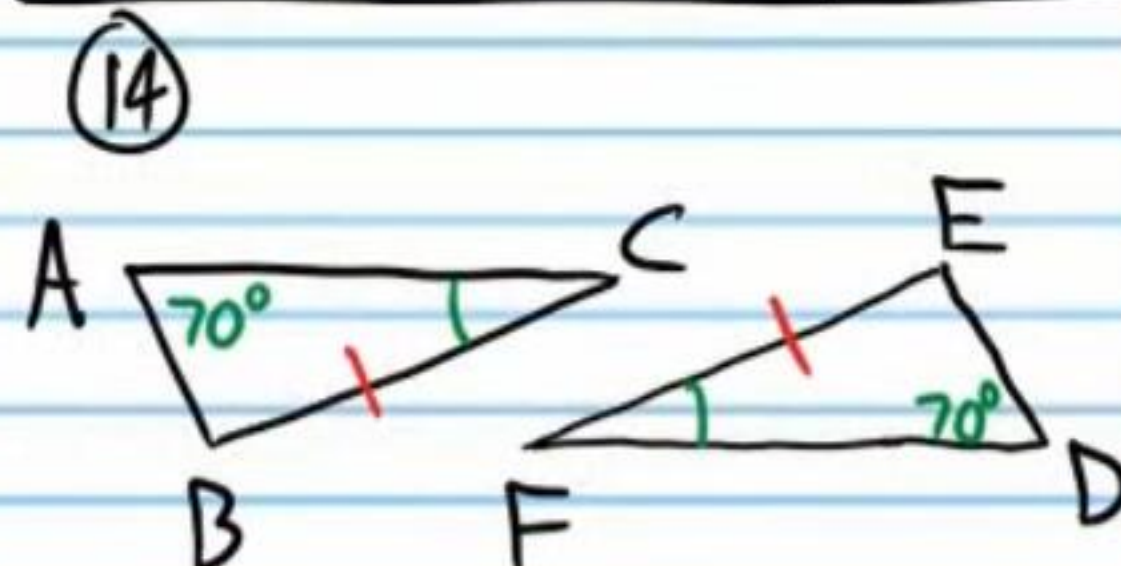
$\triangle ABC \cong \triangle DEF$ by SSS



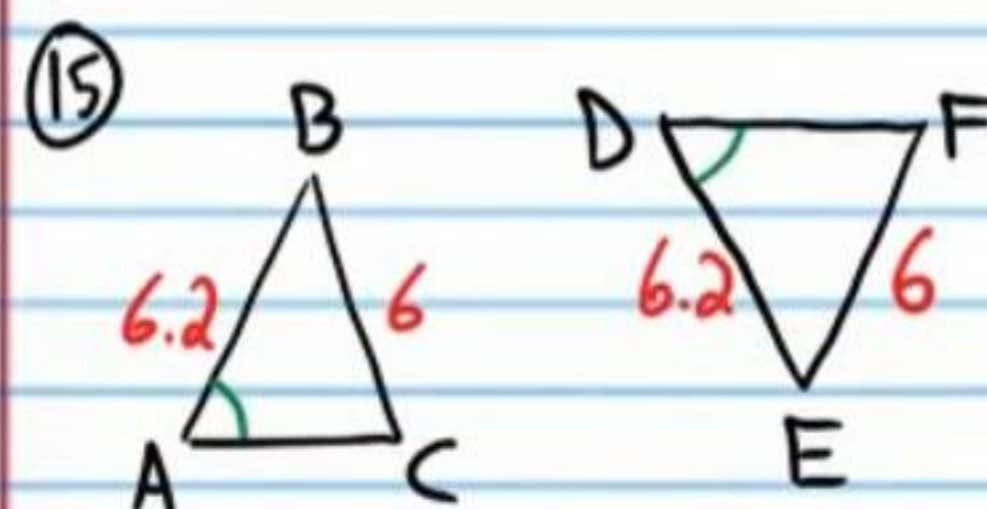
$\triangle ABC \cong \triangle DEF$ by SAS



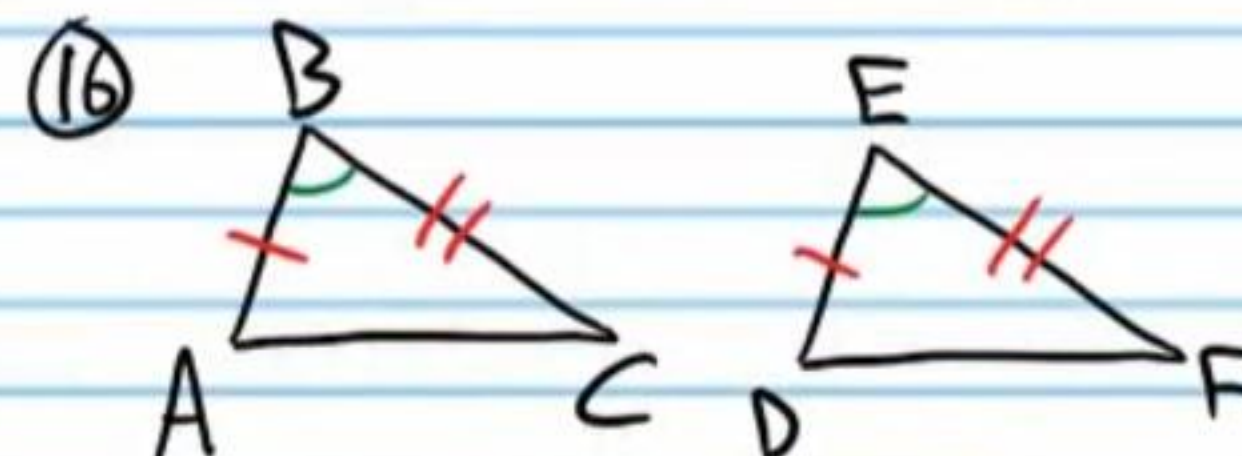
$\triangle ABC \cong \triangle DEF$ by ASA



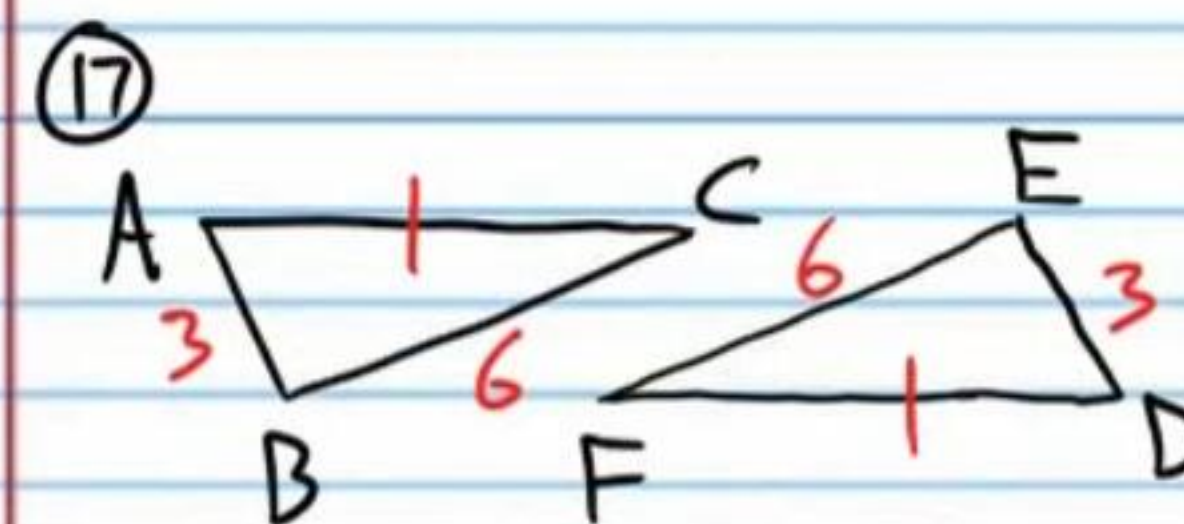
$\triangle ABC \cong \triangle DEF$ by AAS



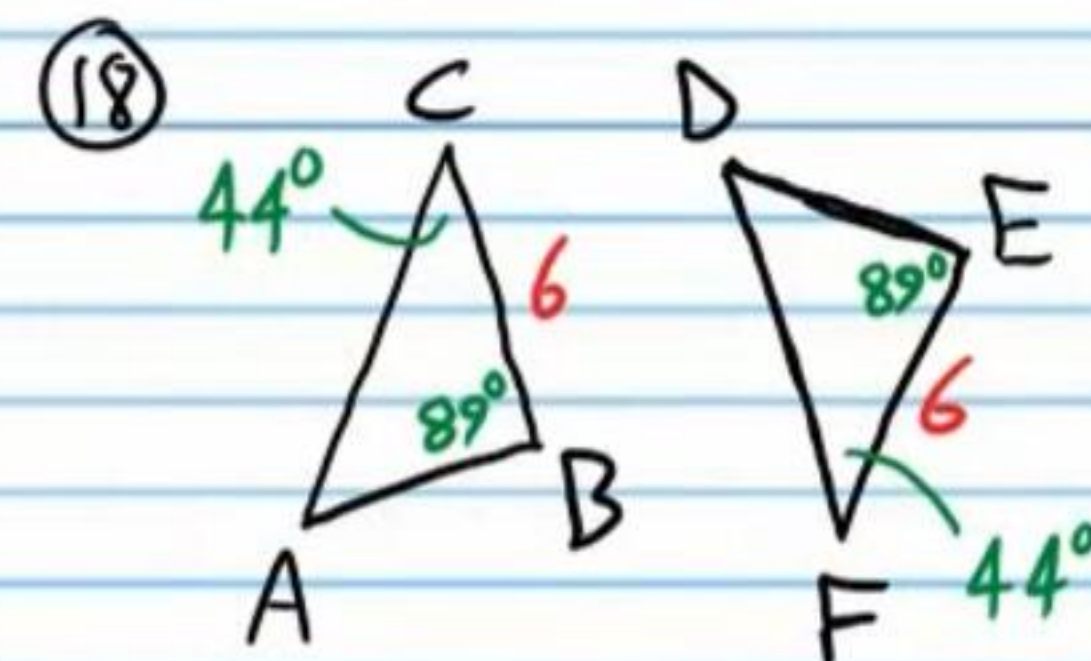
Not enough information



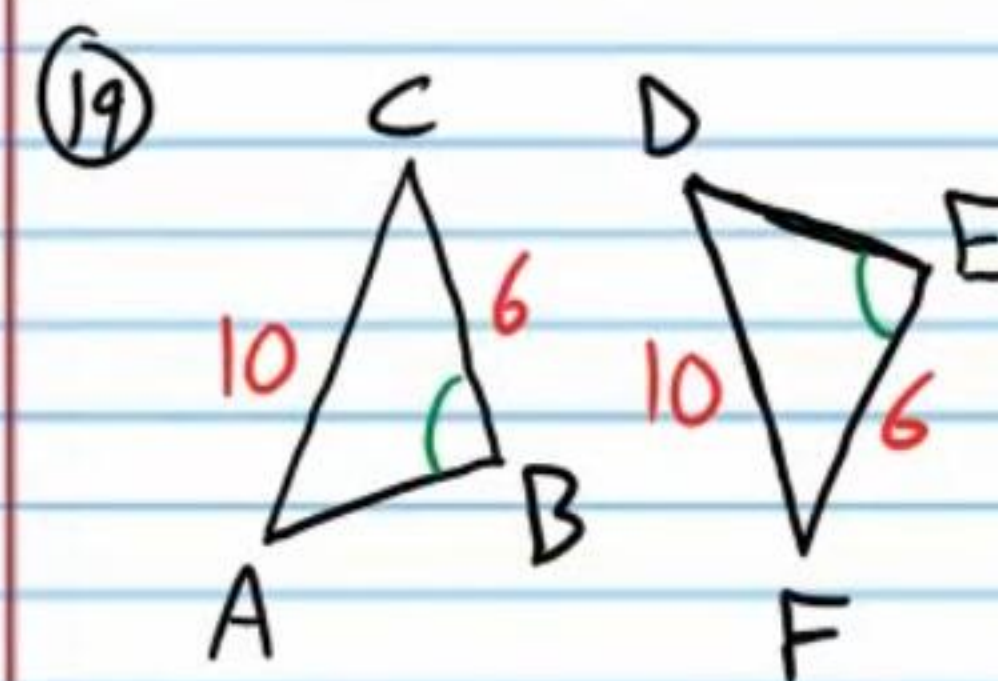
$\triangle ABC \cong \triangle DEF$ by SAS



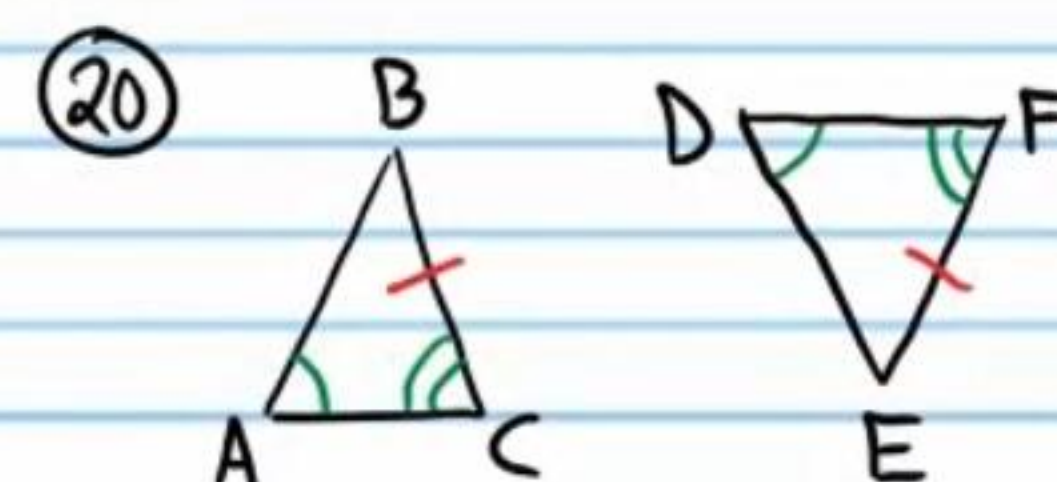
$\triangle ABC \cong \triangle DEF$ by SSS



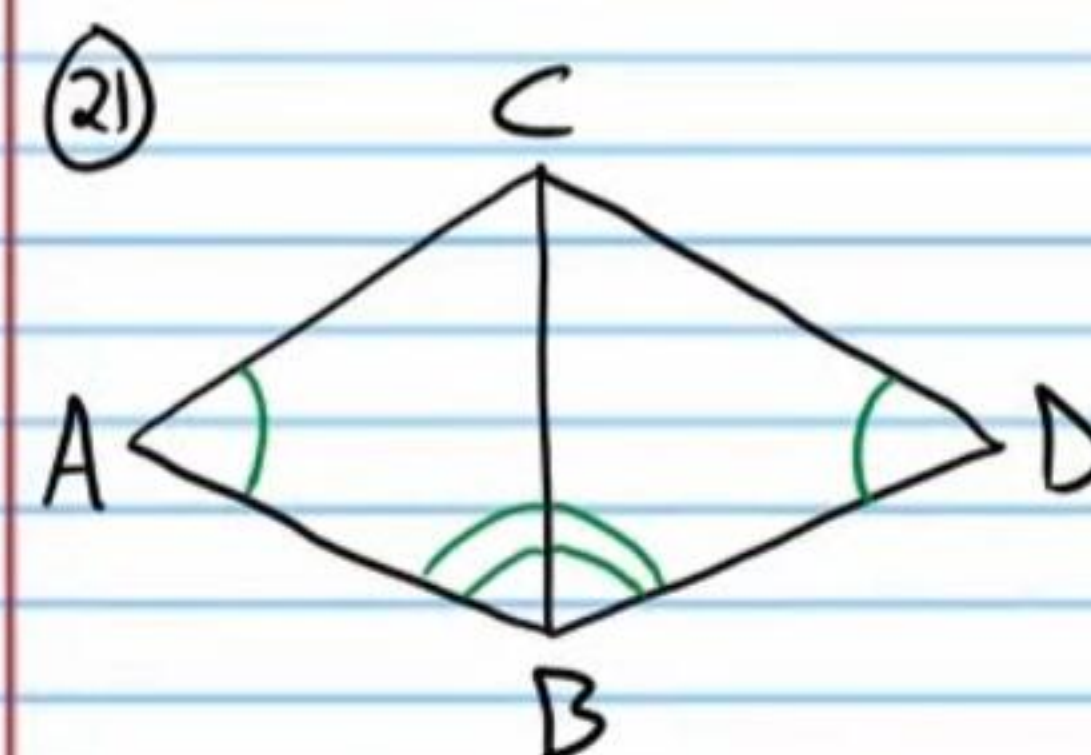
$\triangle ABC \cong \triangle DEF$ by ASA



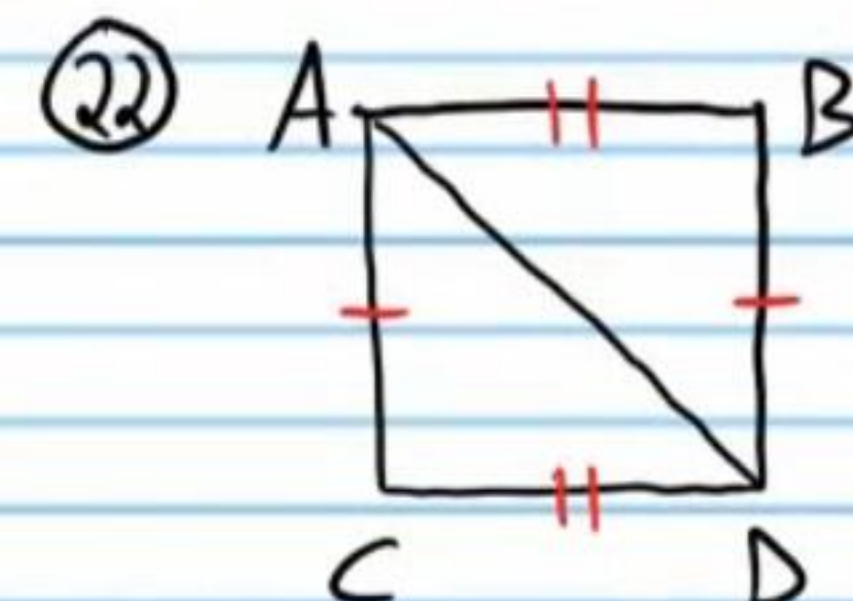
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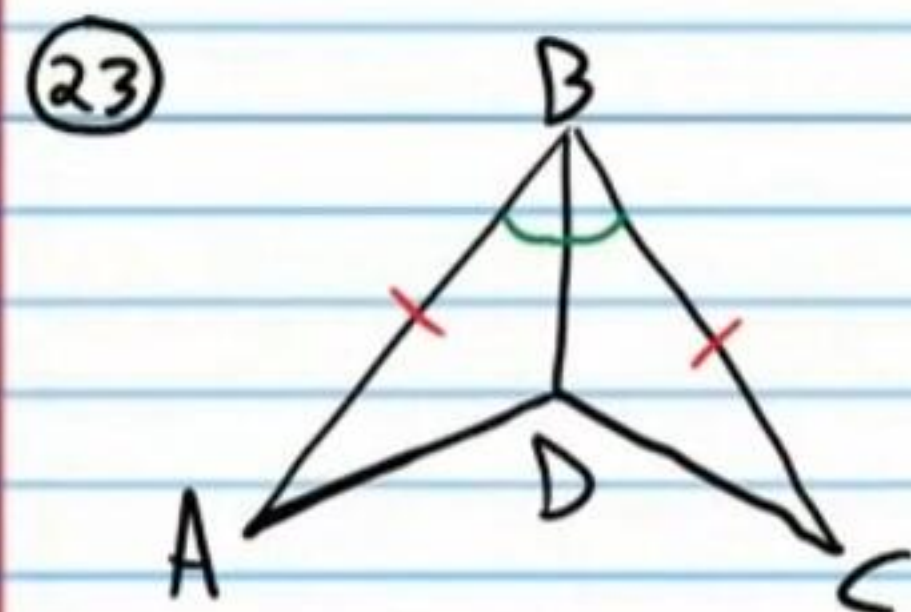
$\triangle ABC \cong \triangle DEF$ by AAS



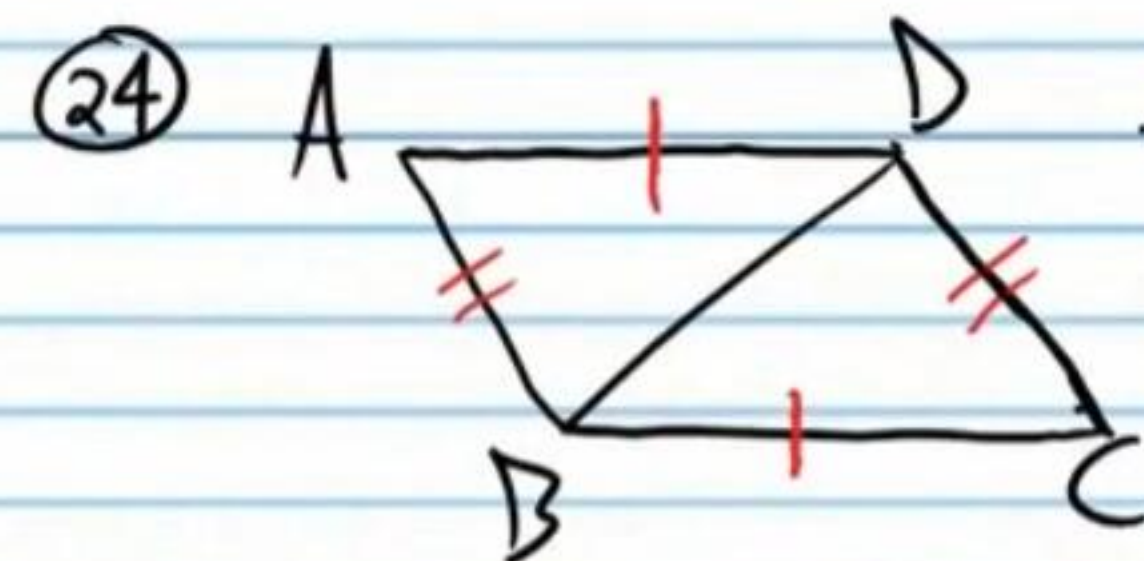
$\triangle ABC \cong \triangle DCB$ by AAS



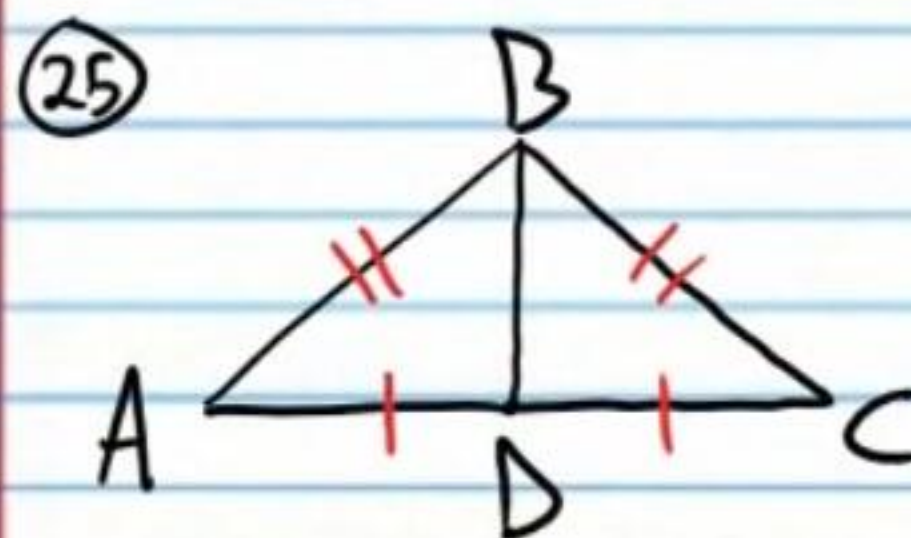
$\triangle ABD \cong \triangle CDB$ by SSS



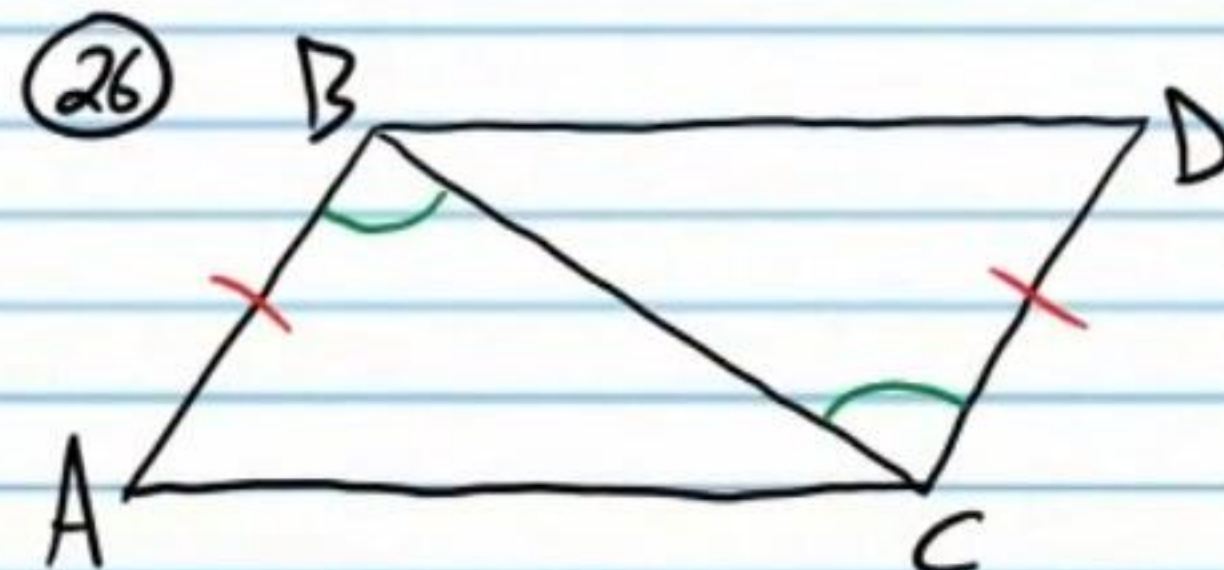
$\triangle ABD \cong \triangle CBD$ by SAS



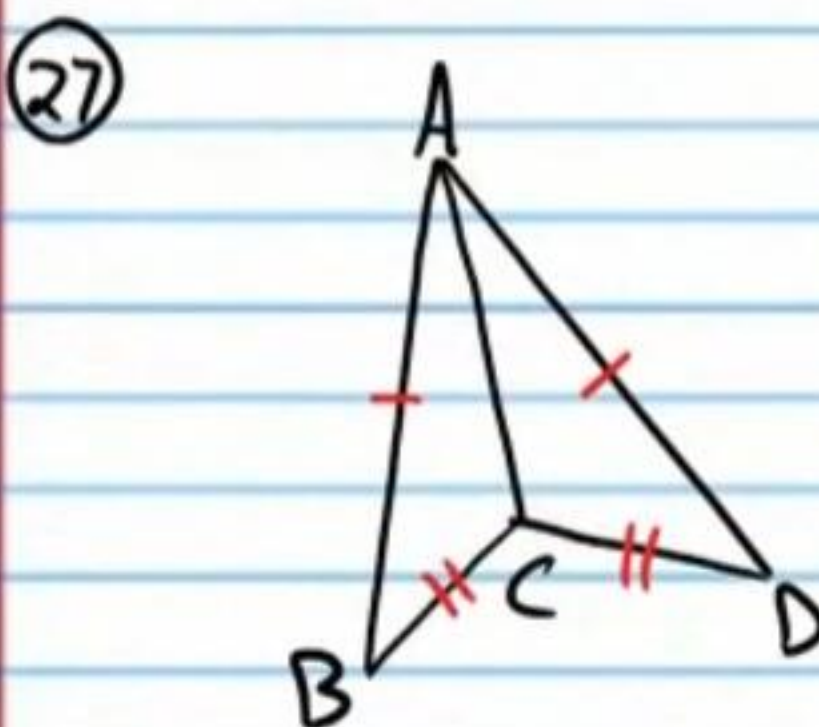
$\triangle ABD \cong \triangle CDB$ by SSS



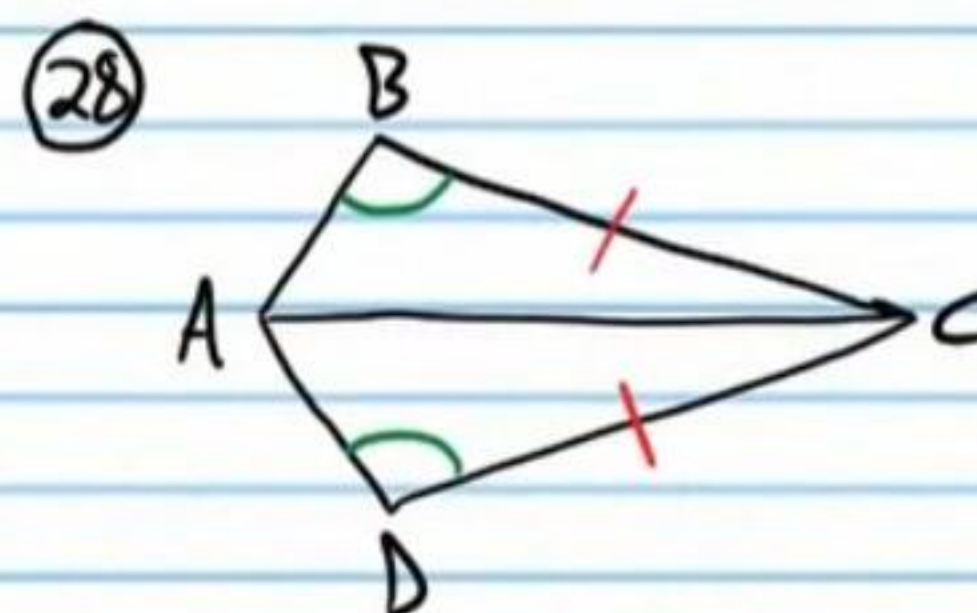
$\triangle ABD \cong \triangle CBD$ by SSS



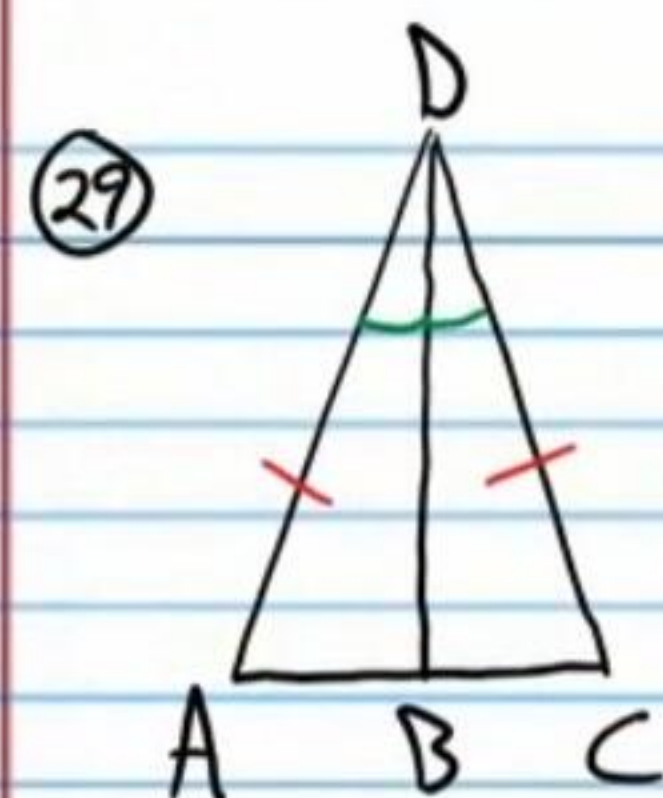
$\triangle ABC \cong \triangle DCB$ by SAS



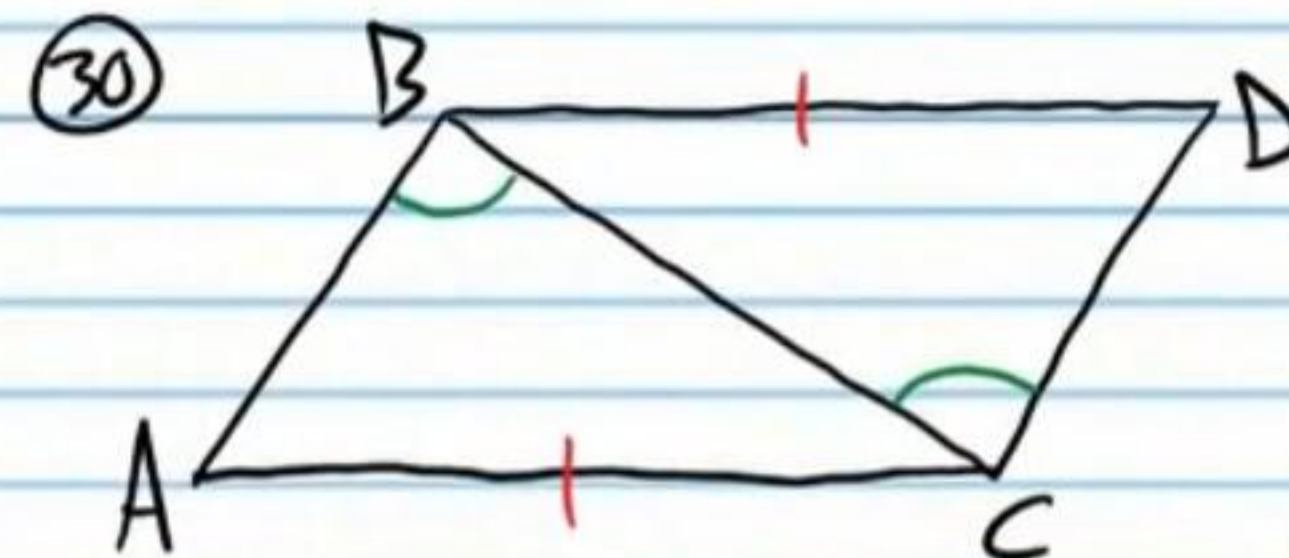
$\triangle ABC \cong \triangle ADC$ by SSS



Not enough information.

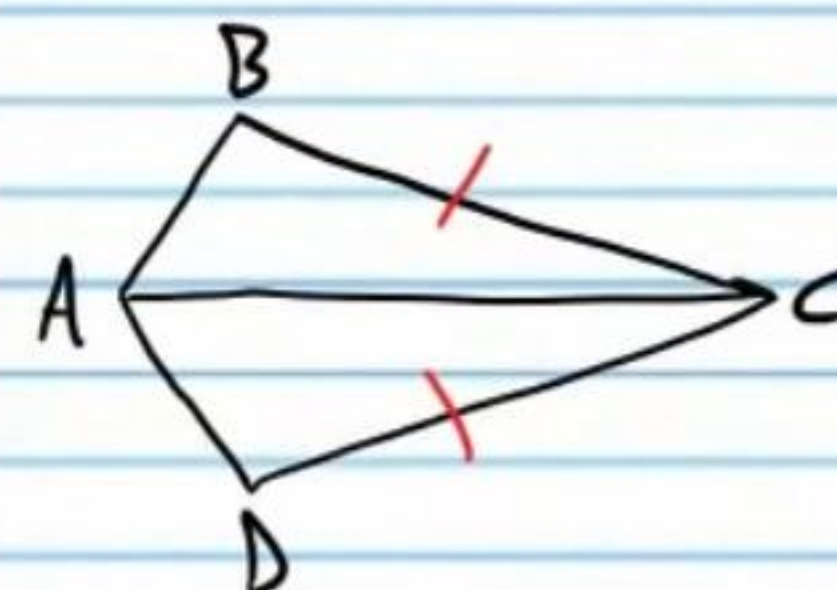


$\triangle ABD \cong \triangle CBD$ by SAS



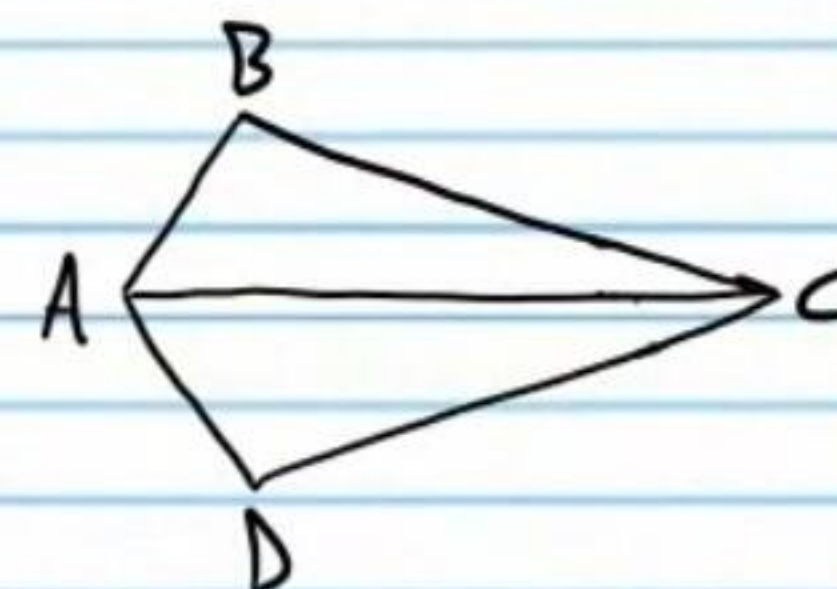
Not enough information

31) $\overline{AC}$ bisects $\angle BCD$.

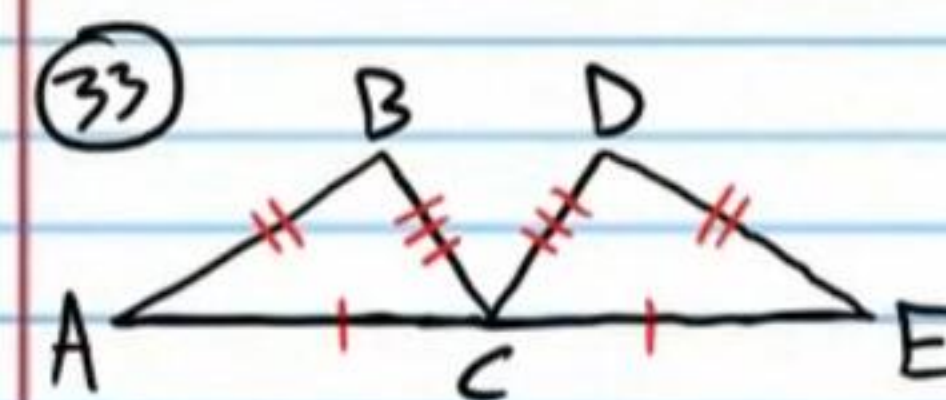


$\triangle ABC \cong \triangle ADC$ by SAS

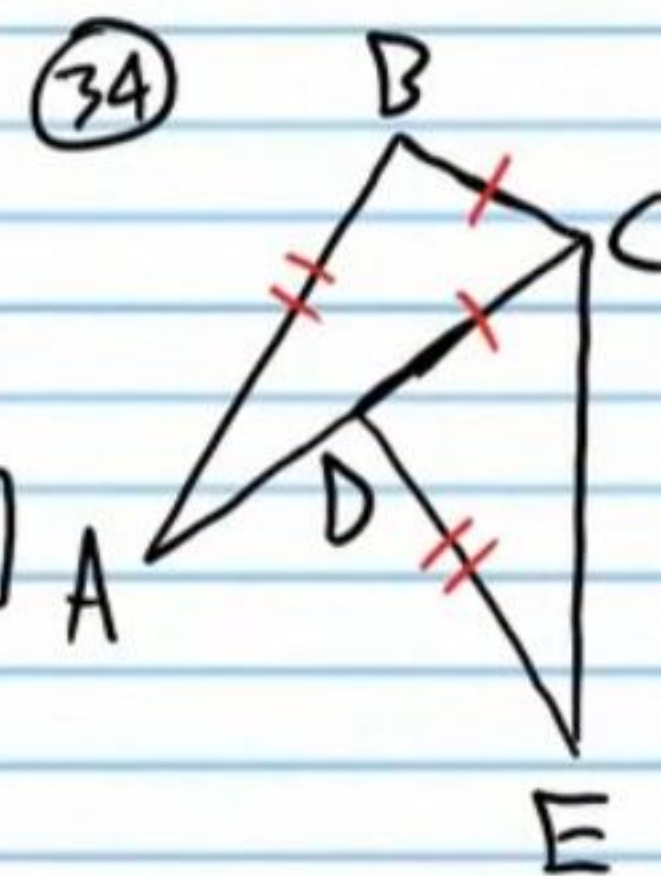
32) $\overline{AC}$ bisects $\angle DAB$ and also $\angle BCD$.



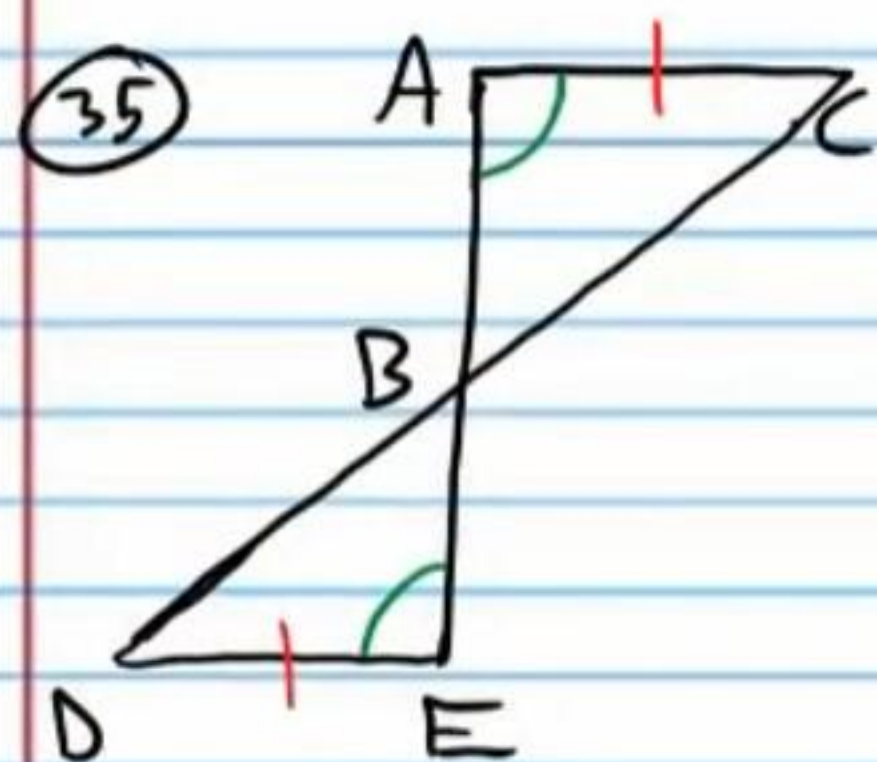
$\triangle ABC \cong \triangle ADC$ by ASA



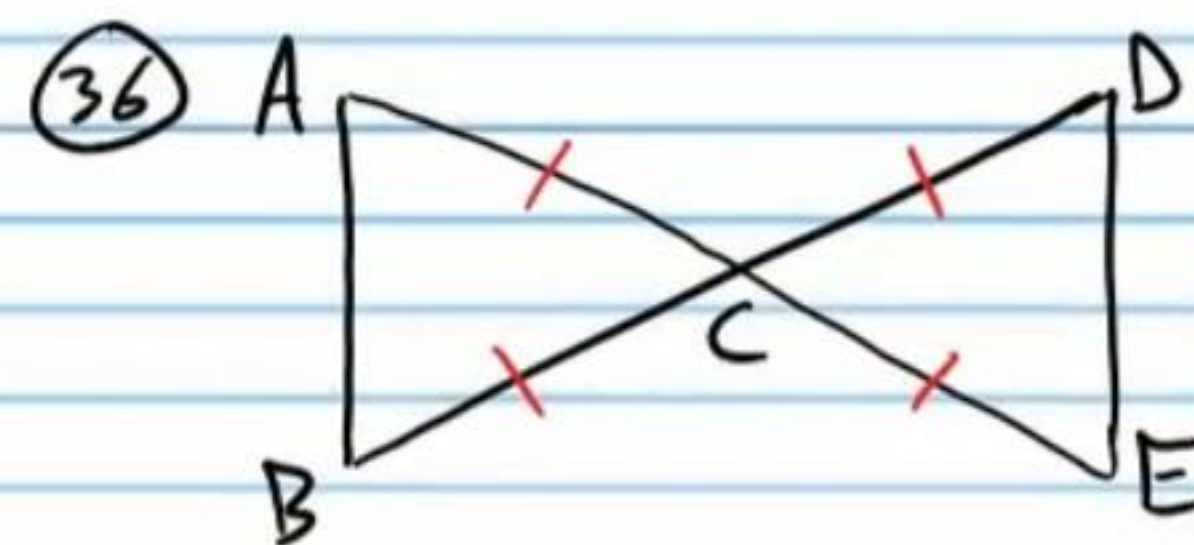
$\triangle ABC \cong \triangle EDC$ by SSS



Not enough information

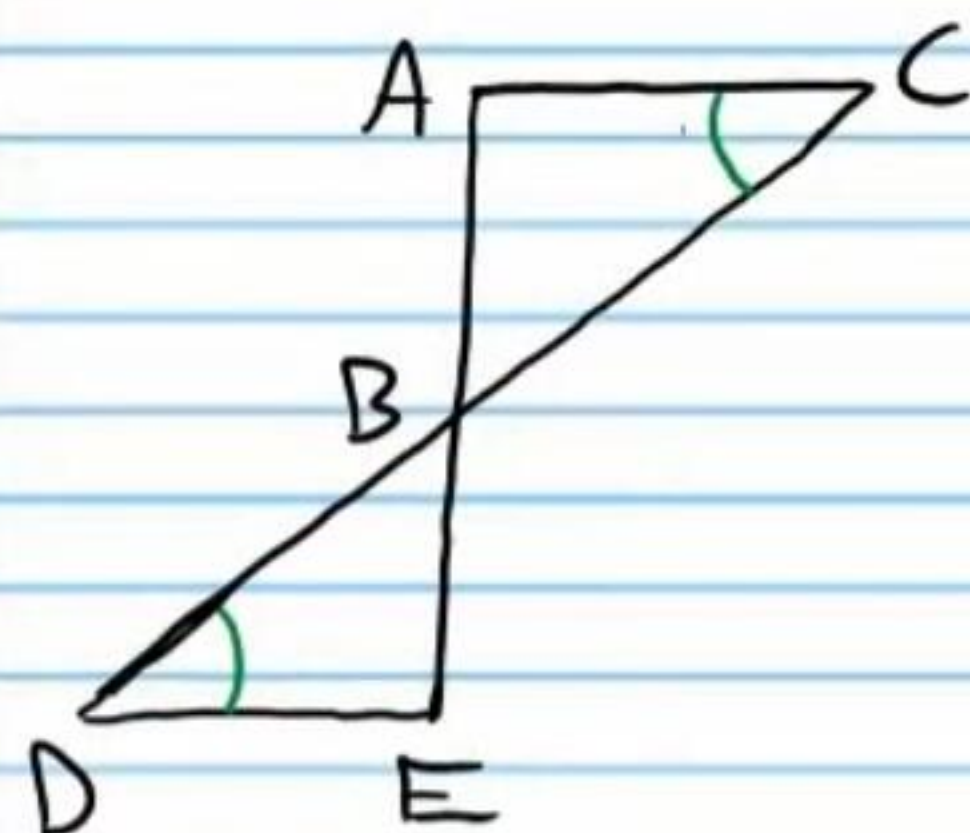


$\triangle ABC \cong \triangle EBD$ by AAS



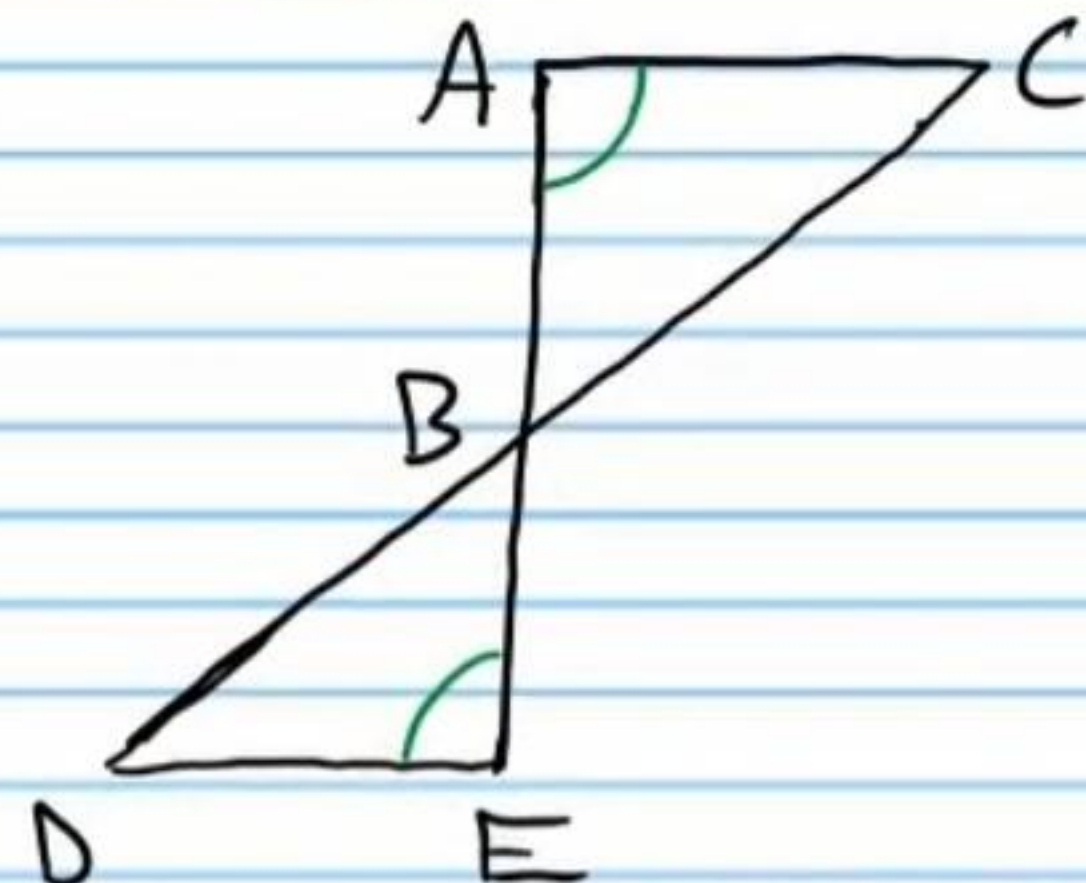
$\triangle ACB \cong \triangle ECD$ by SAS

37 B is the midpoint of $\overline{DC}$.



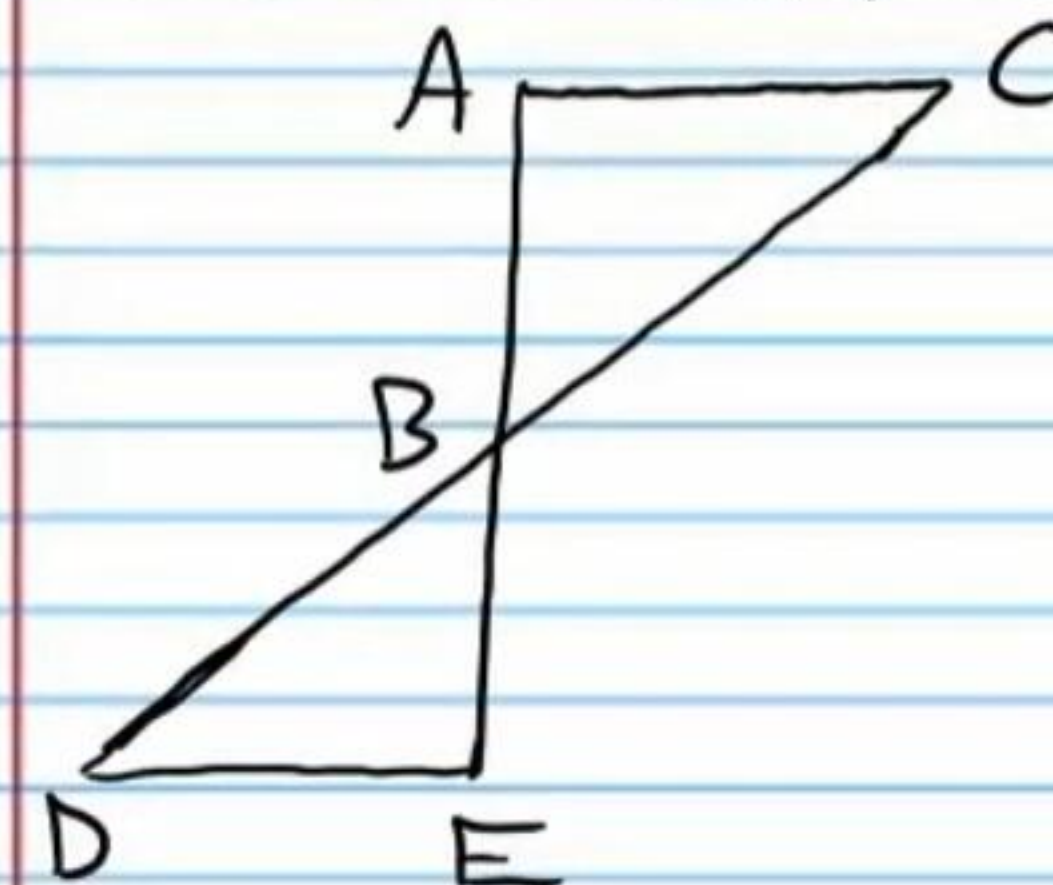
$\triangle ABC \cong \triangle EBD$ by ASA

38 B is the midpoint of $\overline{CD}$.



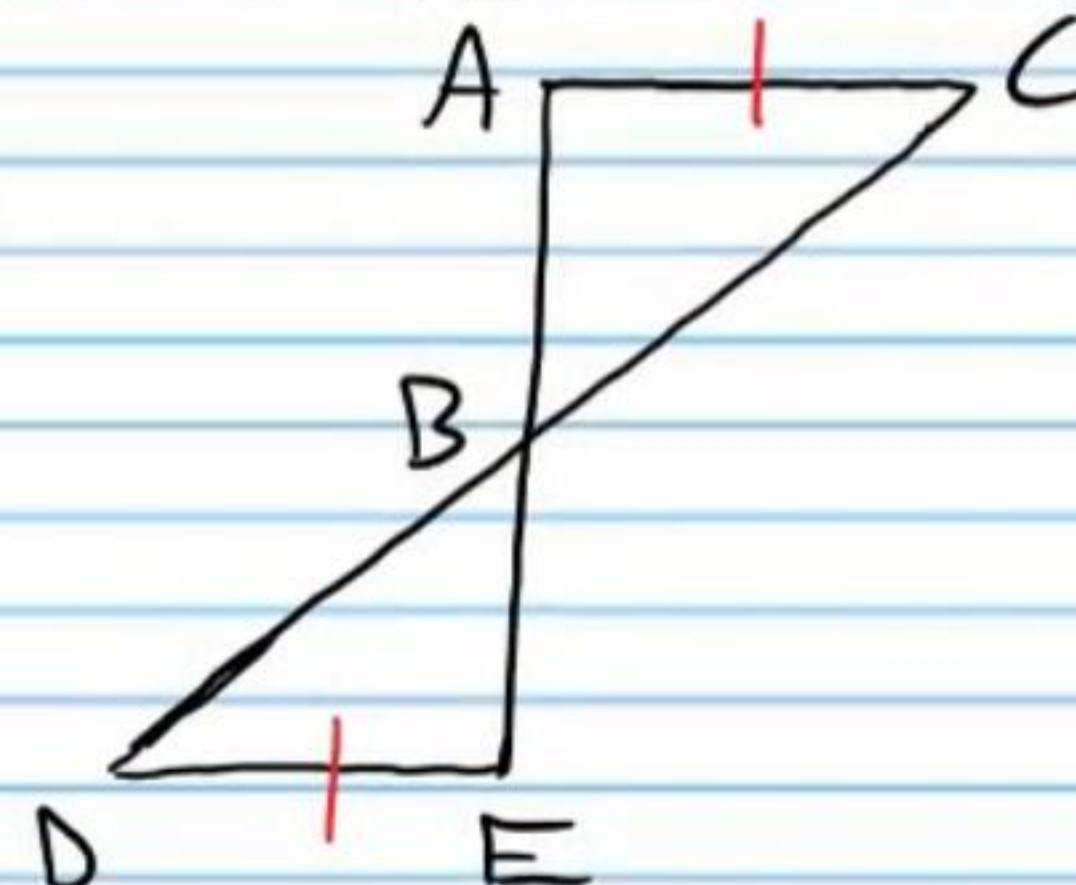
$\triangle ABC \cong \triangle EBD$ by AAS

39 B is the midpoint of $\overline{CD}$ and B is also the midpoint of $\overline{AE}$.



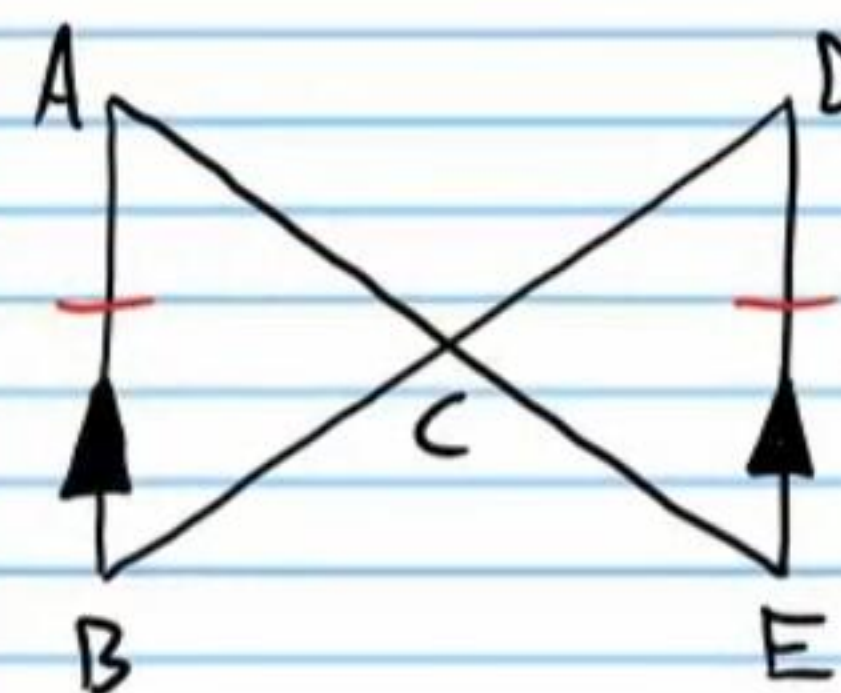
$\triangle ABC \cong \triangle EBD$ by SAS

40 B is the midpoint of $\overline{AE}$.



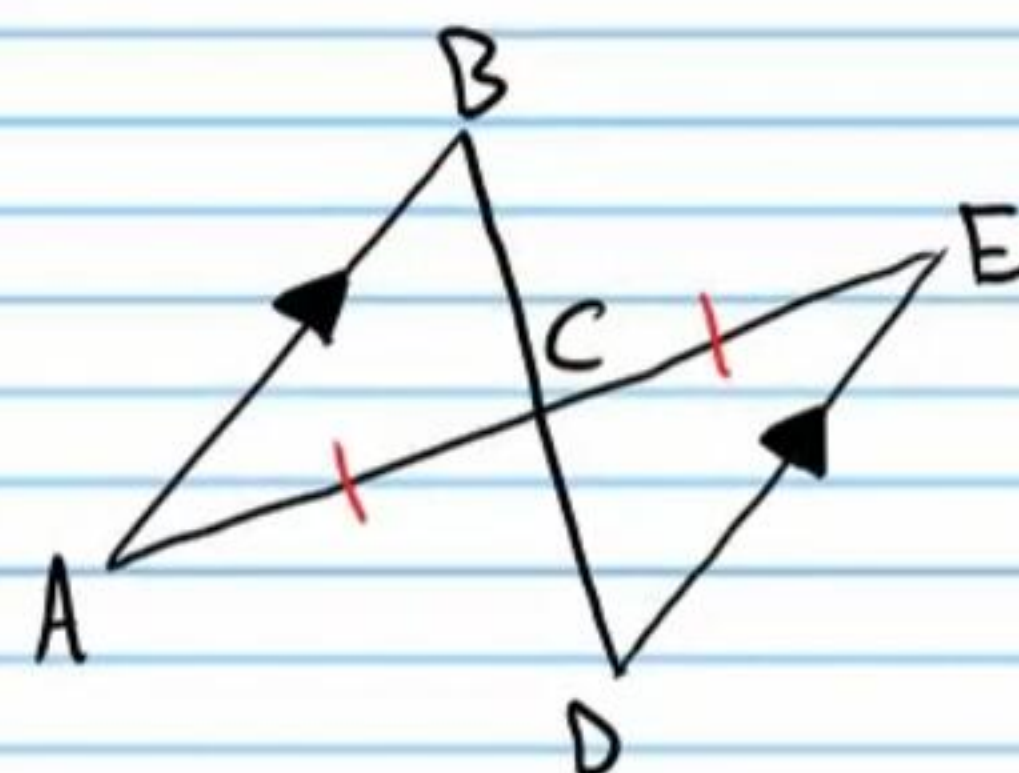
Not enough information

41 $\overline{AB} \parallel \overline{DE}$.



$\triangle ABC \cong \triangle EDC$ by AAS

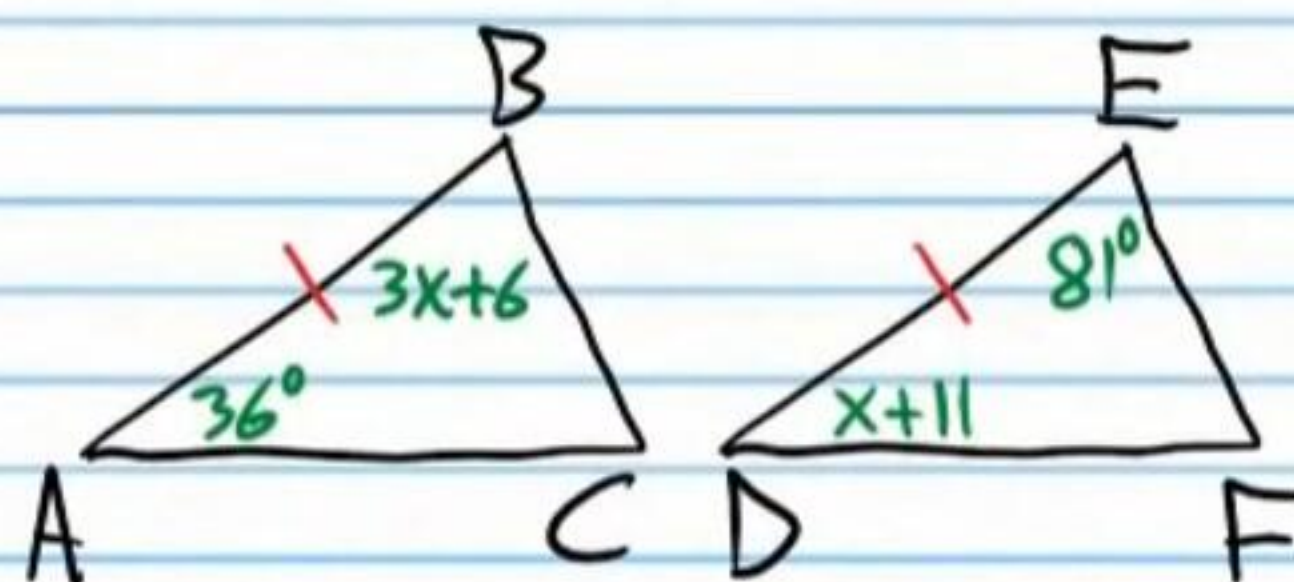
42 $\overline{AB} \parallel \overline{DE}$.



$\triangle ABC \cong \triangle EDC$ by ASA

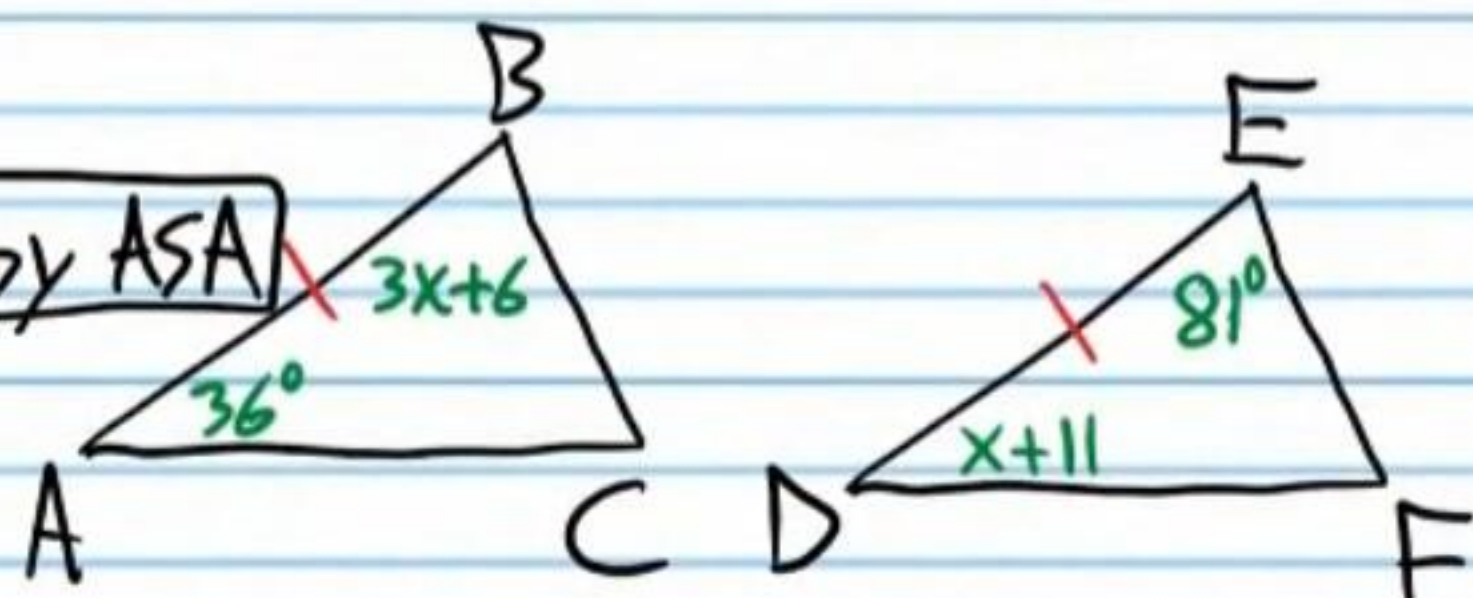
④③ $x = 20$.

Not $\cong$

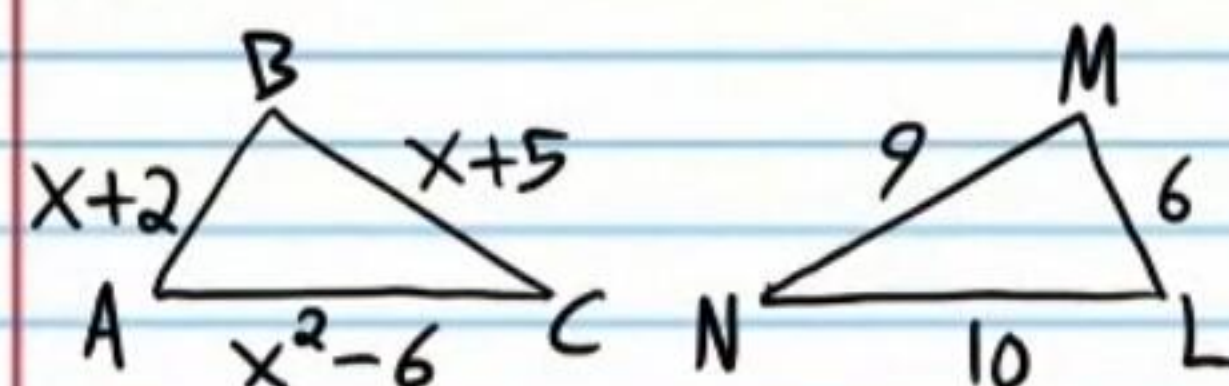


④④ $x = 25$.

$\triangle ABC \cong \triangle DEF$ by ASA

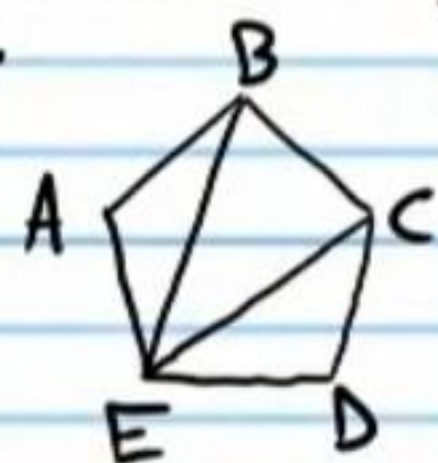


④⑤ $x = 4$.



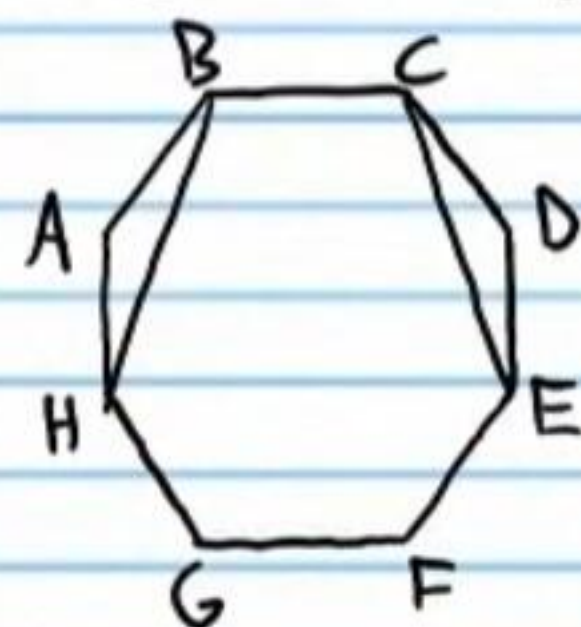
$\triangle ABC \cong \triangle LMN$ by SSS

④⑥ ABCDE is a regular pentagon.



$\triangle ABE \cong \triangle DEC$ by SAS

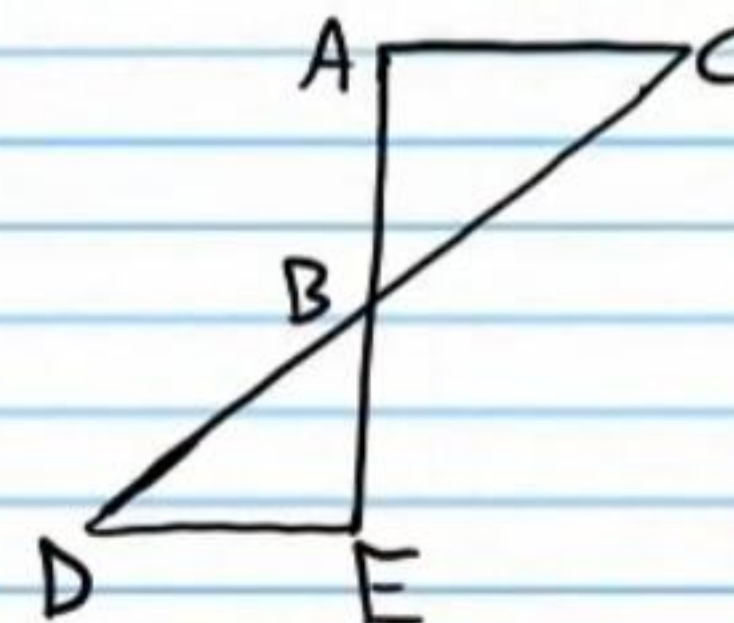
④⑦ ABCDEFGH is a regular octagon.



$\triangle CDE \cong \triangle BAH$ by SAS

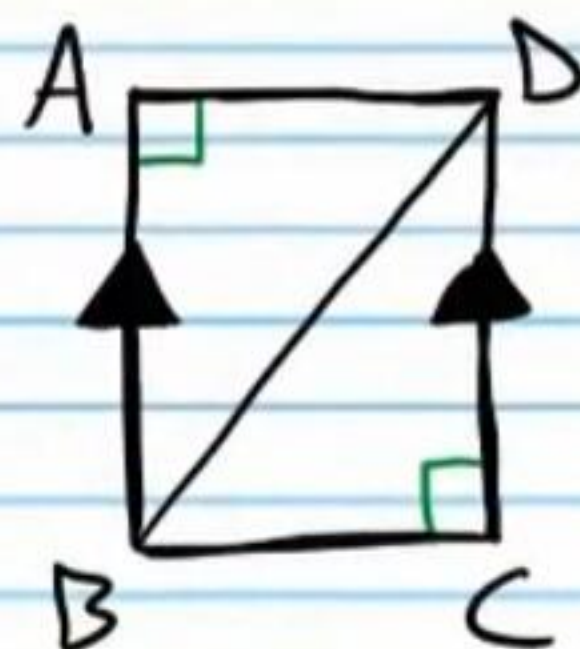
Prove the following statements using two-column proofs.

④⑧ If B bisects $\overline{AE}$, and $\angle BAC$ and $\angle DEB$ are right angles, then $\triangle ABC \cong \triangle EBD$.



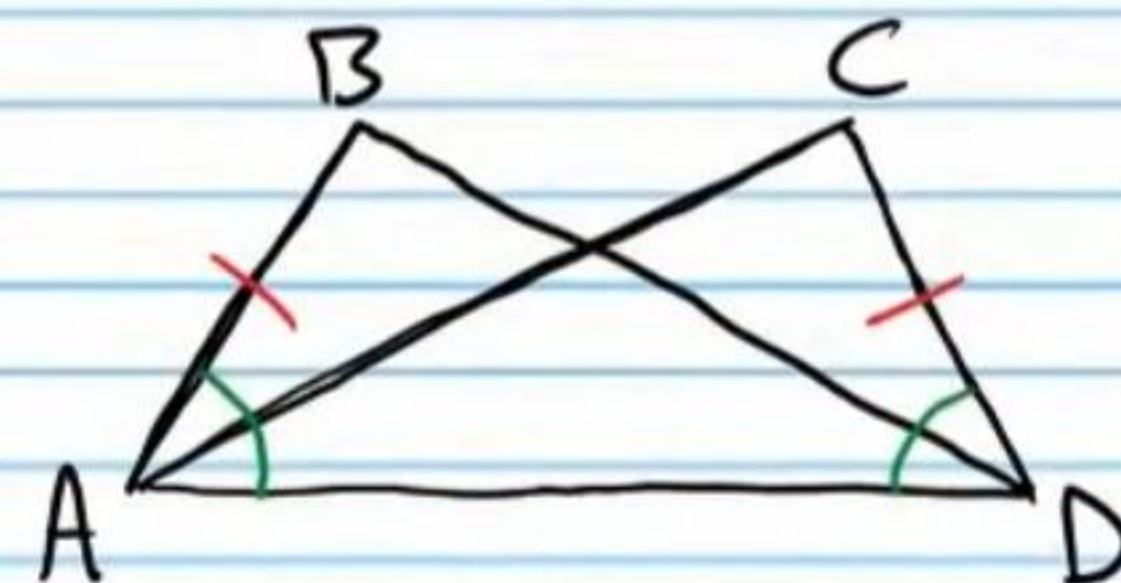
Statement	Reason
I) B bisects $\overline{AE}$.	Given.
II) $\overline{AB} \cong \overline{BE}$	Def. of seg bisector.
III) $\angle BAC$ and $\angle DEB$ are Rt. $\angle$'s	Given.
IV) $\angle BAC \cong \angle DEB$	Right $\angle \cong$ Thm.
V) $\angle ABC \cong \angle DBE$	Vert. $\angle$'s Thm
VI) $\triangle ABC \cong \triangle EBD$	ASA

④9 If $\angle A$ and $\angle C$ are both right angles, and if $\overline{AB} \parallel \overline{DC}$, then $\triangle ABD \cong \triangle CDB$.



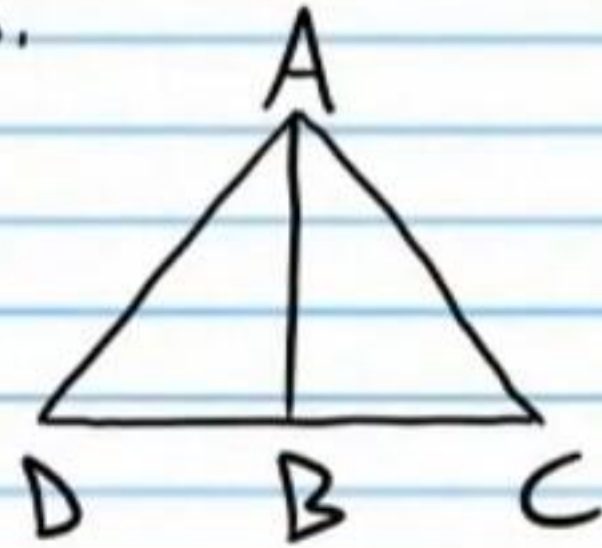
Statement	Reason
I) $\angle A$ & $\angle C$ are rt $\angle$'s.	Given
II) $\angle A \cong \angle C$	Right $\angle \cong$ Thm.
III) $\angle ABD \cong \angle CDB$	Alt. Int $\angle$'s Thm.
IV) $\overline{BD} \cong \overline{BD}$	Ref. Prop of $\cong$
V) $\triangle ABD \cong \triangle CDB$	AAS

⑤0 If $\angle BAD \cong \angle CDA$ and $\overline{AB} \cong \overline{CD}$ in the figure below, then $\triangle ABD \cong \triangle DCA$.



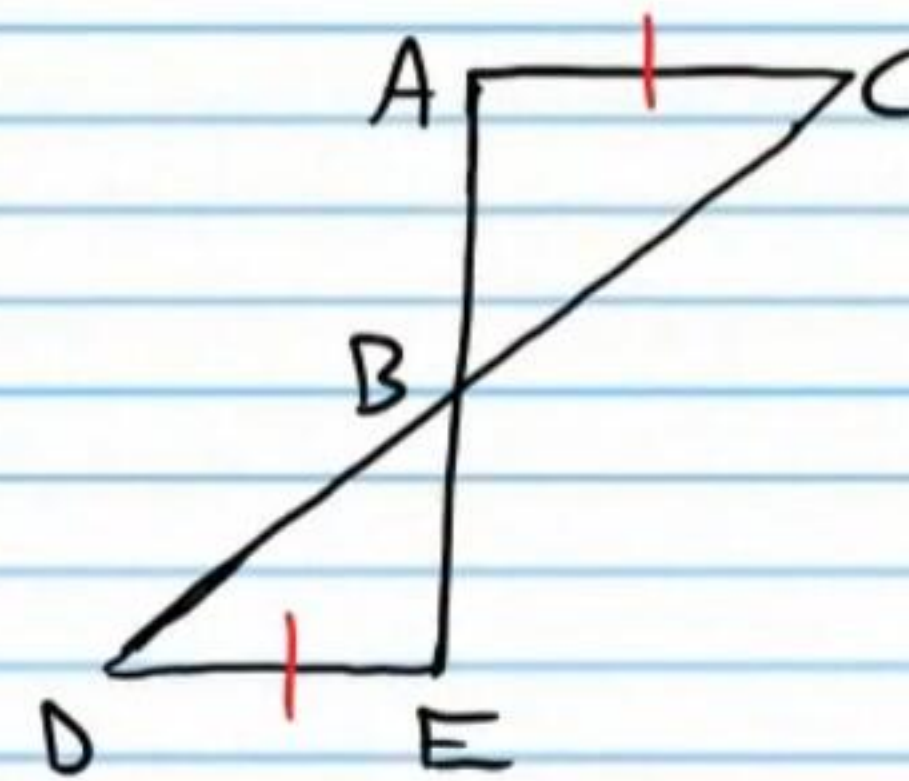
Statement	Reason
I) $\angle BAD \cong \angle CDA$	Given.
II) $\overline{AB} \cong \overline{CD}$	Given.
III) $\overline{AD} \cong \overline{AD}$	Ref. Prop of $\cong$.
IV) $\triangle ABD \cong \triangle DCA$	SAS

⑤) If $\overline{AB}$ is perpendicular to $\overline{DC}$ in the figure below, and B is the midpoint of $\overline{DC}$, then $\triangle DBA \cong \triangle CBA$.



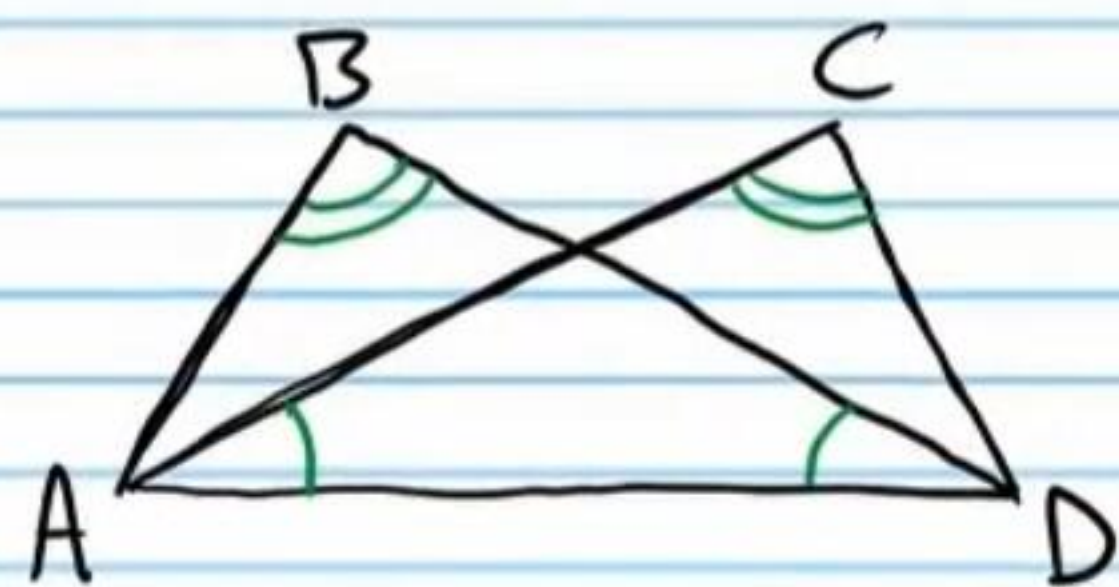
Statement	Reason
I) $\overline{AB} \perp \overline{DC}$	Given.
II) $\angle DBA$ & $\angle CBA$ are rt $\angle$'s.	Def. of $\perp$.
III) $\angle DBA \cong \angle CBA$.	Right $\angle \cong$ Thm.
IV) B is the midpoint of DC.	Given.
V) $\overline{DB} \cong \overline{BC}$	Def. of midpoint.
VI) $\overline{AB} \cong \overline{AB}$	Ref. Prop of $\cong$
VII) $\triangle DBA \cong \triangle CBA$	SAS

⑤a) If $\overline{AC} \parallel \overline{DE}$ and $\overline{AC} \cong \overline{DE}$, then $\triangle ABC \cong \triangle EBD$.



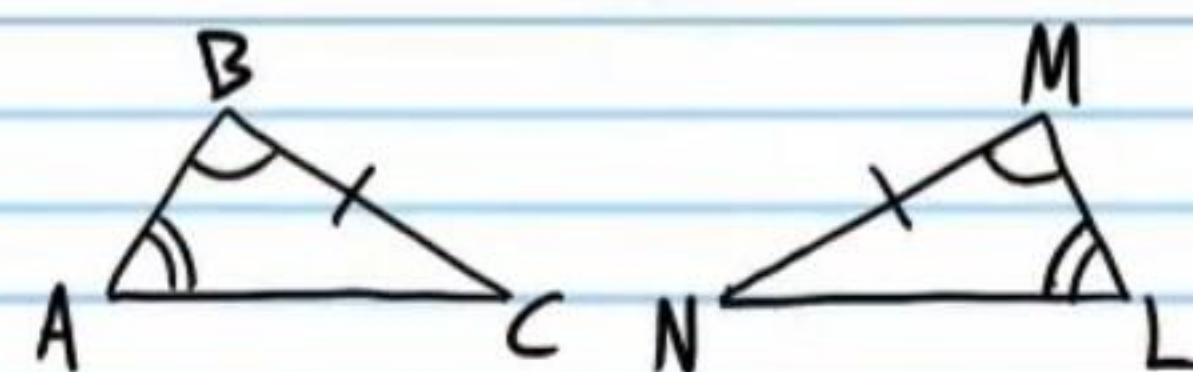
Statement	Reason
I) $\overline{AC} \parallel \overline{DE}$	Given
II) $\angle ACB \cong \angle BDE$.	Alt. Int. $\angle$'s Thm.
III) $\angle ABC \cong \angle EBD$	Vert. $\angle$'s Thm.
IV) $\overline{AC} \cong \overline{DE}$	Given (diagram)
V) $\triangle ABC \cong \triangle EBD$	AAS

⑤③ If $\angle CAD \cong \angle BDA$ and $\angle ABD \cong \angle DCA$, then $\triangle ABD \cong \triangle DCA$.



Statement	Reason
I) $\angle CAD \cong \angle BDA$ and $\angle ABD \cong \angle DCA$	Given.
II) $\overline{AD} \cong \overline{AD}$	Ref. Prop of $\cong$.
III) $\triangle ABD \cong \triangle DCA$	AAS

⑤④ Use the Third Angles Theorem to prove the AAS Congruence Theorem. We proved the Third Angles Theorem in a previous class. It says that if two angles in one triangle are congruent to two angles in another triangle, then the third angles in each triangle must be congruent.



Statement	Reason
I) $\angle ABC \cong \angle LMN$ & $\angle BAC \cong \angle MLN$	Given.
II) $\overline{BC} \cong \overline{MN}$	Given.
III) $\angle ACB \cong \angle LNM$	3 rd $\angle$'s Thm.
IV) $\triangle ABC \cong \triangle LMN$	ASA